INTEGRAL VANISHING CYCLES FROM PERIODS ON CONIFOLD SMOOTHINGS

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Abstract. We give a terminating construction of the marked integral vanishing-cycle map for a six-parameter family of Calabi–Yau conifold smoothings in $\mathbb{P}^3 \times \mathbb{P}^3$. The eight vanishing spheres span a primitive rank-six lattice whose symplectic normal form is determined by topology. We recover their coordinates in an independent geometric marking by constructing an effective neighbourhood in vanishing-period coordinates and absolute error bounds for cycle transport and integration. Picard–Lefschetz period jumps determine the integral columns and monodromy without an a priori bound on the integer entries or restrictions on nonzero period ratios. Recovery at one reference fibre suffices throughout the smooth locus. As applications, we verify the hypotheses of established analytic results on the same fibres in a possibly ineffective smaller neighbourhood. Over $\mathbb{R}$, the kernel of the recovered map is the image of divisor restriction to the exceptional curves; its capacity-weighted quadratic form and limiting harmonic mass give relative asymptotics for the two small positive Ricci-flat two-form eigenvalues. For the constructed nonflat polystable rank-four bundles, a local invariant manifold of Hermitian–Yang–Mills metric flow is tangent at the Hermitian–Einstein metric to the complete low spectral subspace of the Laplacian on trace-free Hermitian endomorphisms, including its kernel.

1. Introduction

A conifold transition relates exceptional curves on a small resolution to vanishing three-spheres on a smoothing. The relations among those spheres give only the kernel of the homomorphism sending each sphere to its class in middle homology. The integral placement of the classes themselves enters the integral monodromy representation and the saturation of the vanishing-cycle lattice. For the compact family studied here, integral topology shows that the lattice is primitive and determines this placement up to an integral symplectic change of basis. The coordinates of the spheres in an independently constructed geometric marking remain to be identified. We recover those coordinates and the corresponding monodromy matrices from geometrically represented period observations on a family of Calabi–Yau threefolds.

The family consists of complete intersections of bidegrees $(1,1)$, $(1,1)$ and $(2,2)$ in $\mathbb{P}^3 \times \mathbb{P}^3$, with polarization $\mathcal{O}(2,2)$. Its central fibre has eight ordinary double points, divided into two groups I_f, I_g of four. The projective small resolution has second Betti number four; smooth fibres have second Betti number two and free middle homology of rank 134. Orient the local vanishing spheres δ_i so that their only integral relations are

$$\sum_{i \in I_f} \delta_i = 0, \qquad \sum_{i \in I_g} \delta_i = 0. \tag{1}$$

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These two relations are supplied as geometric data. The unknown is the marked inclusion of their rank-six span in middle homology. Its saturation and torsion-free quotient are determined independently in Proposition 3.2; Corollary 7.2 gives its symplectic normal form.

1.1. **The principal result.** Let

$$R = \{p \in \mathbb{C}^8 : \sum_{i \in I_f} p_i = \sum_{i \in I_g} p_i = 0\}.$$

The relative residue form is denoted by Ψ. An evaluation name for a parameter supplies rational complex approximations to any requested absolute error. This is the meaning of access to an arbitrary complex parameter below; computable parameters give ordinary algorithms.

Theorem 1.1. *There is a terminating selection of algebraic central coefficients and six algebraic coefficient directions with the following properties. One can construct a rational-radius ball $B_{\mathrm{ev}} \subset R$ and a holomorphic coefficient section defining a flat projective family X_p with*

$$\int_{\delta_i} \Psi_p = p_i, \qquad B^{\circ}_{\mathrm{ev}} = B_{\mathrm{ev}} \setminus \bigcup_i \{p_i = 0\}.$$

This is exactly its smooth-fibre locus. The coefficient section, local normal forms and residue units admit evaluation and bounded derivatives of every prescribed finite order on specified smaller domains.

For every named $p \in B^{\circ}_{\mathrm{ev}}$, a terminating procedure constructs a geometric symplectic marking $\gamma_1, \dots, \gamma_{134}$ and recovers

$$D : \mathbb{Z}^8 \longrightarrow H_3(X_p; \mathbb{Z}), \qquad De_i = \delta_i.$$

It returns the marked columns, their primitive integral image, and the individual monodromy matrices, and recovers the topologically determined quotient $\mathbb{Z}^{128}$. These matrices determine the homology action of every loop in B°_{ev}. The observations are the eight local periods and the 1072 complex jumps of the marked periods around represented individual node meridians. Their absolute enclosures are constructed and refined until a strict integer-separation test succeeds. Neither the coordinates of the vanishing spheres nor the homology coordinates of transported endpoints are inputs to this forward calculation.

The proof occupies Sections 3–7; its final assembly is Theorem 7.3. There is no a priori bound on the integer entries and no comparison condition among the nonzero p_i. Termination is pointwise in the smooth parameter. The result does not give a practical running-time bound near the discriminant. In particular, we do not claim numerical values for the selected coefficients or the attachment matrix in the constructed marking. Its six nonzero Smith invariants are all one by topology. The local spheres are globally flat and the marking is transported from a reference fibre. Thus the matrix is constant under this transport, and recovery at the reference fibre suffices. Direct observations at other fibres remain available with the pointwise error bounds proved below.

The finite inverse step has a simple form. If $\delta_i = \sum_b (d_i)_b \gamma_b$ and $J_{ab} = \langle \gamma_a, \gamma_b \rangle$, a positive individual meridian gives

$$\Delta_{ai} = \int_{(F_i)_{\#}\gamma_a} \Psi_p - \int_{\gamma_a} \Psi_p = p_i (J d_i)_a. \tag{2}$$

Thus one holomorphic form suffices for these rank-one variations; its full period functional need not be injective on integral homology. Unimodularity of J recovers d_i from the integers

$(Jd_i)_a$. The substantive construction is the forward access to (2), with valid absolute errors and without knowing those integers.

1.2. **Geometric construction and error control.** We first select an algebraic member of the toric eight-node construction by finite ideal and nonvanishing tests. Algebraic jets at the nodes give the rank-six derivative of its period map and a transverse coefficient slice. Quantitative implicit functions, Morse coordinates and an explicitly integrated residue expansion then give an inverse period section on a known neighbourhood. The analytic estimates include bounds on a larger complex domain, so that differentiation and integration retain controlled errors.

An algebraic smooth reference fibre provides the integral marking. Semialgebraic triangulation constructs its chain complex and integral homology. Face-compatible smoothing gives actual finite piecewise smooth representatives, with first and second derivative bounds. Equation-based horizontal transport moves these representatives to the target and around each individual meridian. The first-jet equation controls the transported tangent vectors, and quadrature of the pulled-back residue form gives absolute enclosures for the period differences.

The denominator p_i may be arbitrarily small. Lemma 7.1 bounds the unknown integer height from the observed numerator and denominator enclosures, then tests whether the ratio error is strictly below $1/2$. Refinement eventually succeeds at every smooth target. The same calculation gives a sufficient output precision depending on the local period and true integer height. The latter enters the explanatory bound, not the adaptive input.

1.3. **Applications on the same fibres.** The spectral restriction map and the recovered attachment map belong to one exact sequence. Writing V_{lost} for the two-dimensional quotient of divisor classes by the surviving classes, the curve restriction map is

$$\mathsf{r} : V_{\mathrm{lost}} \longrightarrow \mathbb{R}^8, \qquad \mathsf{r}(x_f, x_g) = x_f v_f + x_g v_g, \qquad \operatorname{im} \mathsf{r} = \ker(D \otimes \mathbb{R}).$$

For the actual Ricci-flat metric, the weak two-form energy is the capacity-weighted norm of this map. The marked image of D contains additional integral information which that real quadratic form does not determine.

The analytic results of [W26c, W26d] apply after their central geometric and elliptic hypotheses have been verified for the selected coefficients. Section 8 performs this verification on a smaller full ball $B_{\mathrm{an}} \subset B_{\mathrm{ev}}$, whose radius need not be computable. Appendix A records the complete statements used in these applications.

Theorem 9.1 identifies the recovered map with the map in the attachment sequence and relates the analytic constructions on these same fibres. The nonflat rank-four bundles have determinant $\mathcal{O}$, $c_2 H = 32$, and small deformations described by two 2×2 matrices, one in each direction between the two multiplicity spaces. On the polystable locus they admit Hermitian–Einstein metrics. The invariant manifold of metric flow on a fixed bundle is tangent there to the complete seven-dimensional low spectral subspace, including the kernel. Its finite Hessian is relatively equivalent to the squared norm of the infinitesimal unitary action on the matrix parameters. Minimizing the deformation norm over this action and the complementary spatial gauge directions gives the harmonic metric on the varying-bundle moduli space. The proof identifies these maps and quadratic forms using the estimates of [W26d].

The integral observations use represented middle cycles and remain independent of the bundle metric. Their paths lie in B_{ev}° and may leave B_{an}. The bounded residue units in $\tau_i = p_i / U_i$ relate the Hodge capacities to the sufficient absolute precision for direct recovery

at a target fibre. Recovering the constant integer matrix at the reference fibre avoids this target-dependent precision requirement.

1.4. **Relation to previous work.** The Picard–Lefschetz formula and its integral topological interpretation are classical [Del73, Exposé XIV]. Effective recovery of homological monodromy from continued periods is already an explicit part of Lairez, Pichon-Pharabod and Vanhove [LPPV24, §3.5.2, equation (38), arXiv-v2 text]. Their work develops effective homology and periods for projective hypersurfaces and includes topological results for complete intersections in projective space. A product-ambient complete intersection does not become such a complete intersection merely by the Segre embedding. We do not claim the rank-one inversion in (2) as a new method.

Sertöz [Ser19, §§1.3,1.6] gives rigorous period continuation for hypersurfaces and distinguishes transported homology bases from tracking actual representatives. Our construction supplies represented cycles and controlled transport for the present period-coordinate family, including its generally transcendental coefficient section. Lairez–Sertöz [LS23, Theorem 17] instead give an arithmetic separation theorem for periods of algebraic quartic surfaces. Here the quantity to separate is an integer multiple of one nonzero local period with a measured denominator; the stopping argument is elementary once its geometric enclosures have been established.

Vanishing-period coordinates and logarithmic variation in Calabi–Yau conifold transitions are studied by Lee, Lin and Wang [LLW18, §3.3 and Proposition 3.16]. We require, in addition, coefficient selection, a known inverse neighbourhood, a geometric integral marking and absolute observation errors. Map-valued semialgebraic triangulation is an established constructive input [BPR06, Theorem 5.43 and Algorithm 11.44]. The smoothing of its chains used here retains common faces and explicit derivative control; we do not assume a differentiable triangulation with those bounds.

The central compact transition and its divisor data come from [W26a, W26b]. The relative Ricci-flat Hodge theorem and the nonflat bundle, HYM and harmonic quotient results are those of [W26c, W26d]. The present contribution is the terminating geometric observation construction and its integral output on a family to which those analytic conclusions are proved to apply. The final section records the actual connecting maps and an example with independent base and bundle scales.

2. Integral topology and geometric observations

Throughout, a complete intersection has its scheme structure, and its complex orientation fixes Poincaré duality. Put $A = \mathbb{P}^3 \times \mathbb{P}^3$. We first compute the integral topology of every smooth complete intersection of bidegrees $(1,1),(1,1),(2,2)$. This computation does not require its first two equations to cut out a smooth fourfold.

Proposition 2.1. *For any such smooth complete intersection X,*

$$H_1(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = \mathbb{Z}^2, \qquad H_3(X;\mathbb{Z}) = \mathbb{Z}^{134}.$$

The intersection form on $H_3(X;\mathbb{Z})$ is unimodular and alternating.

Proof. Consider the affine coefficient space

$$\mathcal{C} = H^0(A,\mathcal{O}(1,1))^{\oplus 2} \oplus H^0(A,\mathcal{O}(2,2)) \cong \mathbb{C}^{132}. \tag{3}$$

Its locus $\mathcal{U}$ of smooth three-dimensional intersections is Zariski open: the rank-deficient zero locus of the universal equations is closed and projective over $\mathcal{C}$, so its image is closed. A

regular triple has nonempty zero set because its intersection class is nonzero. Successive Bertini for the very ample bundles gives a triple whose entire successive flag is smooth.

The open set $\mathcal{U}$ is path connected. Two of its points can be joined through a third point along complex lines not contained in its algebraic complement; each line meets that complement in finitely many points, which a path can avoid. The universal family over $\mathcal{U}$ is proper and smooth. Transport along the resulting path identifies the oriented integral topology of X with that of a smooth flag; see [Ehr52, §§1,3] for the underlying proper-submersion transport.

On the smooth flag the ambient dimensions are successively $6, 5, 4, 3$. Integral Lefschetz for a smooth very ample divisor therefore preserves H_1 and H_2 at all three steps [Mil63, Corollary 7.3 and Theorem 7.4]. Those groups of A are respectively zero and $\mathbb{Z}^2$, proving the first two assertions. It also proves connectedness. The deformation is used only to compute topology; it does not replace the specific observation family.

All groups are finitely generated. Universal coefficients give

$$0 \longrightarrow \operatorname{Ext}(H_2(X;\mathbb{Z}),\mathbb{Z}) \longrightarrow H^3(X;\mathbb{Z}) \longrightarrow \operatorname{Hom}(H_3(X;\mathbb{Z}),\mathbb{Z}) \longrightarrow 0.$$

The left term vanishes, so H^3 is free. Integral Poincaré duality then makes H_3 free and its pairing unimodular [Hat02, Theorems 3.2, 3.30 and Proposition 3.38]. Graded commutativity in degree three makes the pairing alternating.

For the rank, let x, y be the ambient hyperplane classes and put $s = x + y$. The tangent and normal bundle sequences give

$$[X] = 2s^3, \qquad c(TX) = \frac{(1+x)^4(1+y)^4}{(1+s)^2(1+2s)}, \qquad c_3(TX) = -2(x^3+y^3) - 10(x^2y + xy^2). \tag{4}$$

The coefficient of x^3y^3 in $2s^3c_3(TX)$ is -128. Thus $\chi(X) = -128$. Connectedness and duality give $b_0 = b_6 = 1$, $b_1 = b_5 = 0$, and $b_2 = b_4 = 2$, whence $b_3 = 134$. □

2.1. **Markings and the observation map.**

Definition 2.2. A geometric marking of X consists of an integral basis $\gamma_1, \dots, \gamma_{134}$ of $H_3(X;\mathbb{Z})$, finite piecewise smooth integral singular cycles c_a representing it, and the intersection matrix $J_{ab} = \langle \gamma_a, \gamma_b \rangle$. It is symplectic if J is the chosen standard symplectic matrix.

Such markings exist: a unimodular alternating lattice admits an integral symplectic basis, and smooth singular homology represents its classes by finite smooth chains. Section 5 constructs the marking and derivative bounds from algebraic fibre equations. It does not require the coordinates of any vanishing cycle.

For a smooth proper family along a loop ℓ, choose a smooth horizontal splitting and let $F_\ell : X \longrightarrow X$ be the endpoint transport. For a closed three-form Ψ on X, define

$$\Pi_a = \int_{c_a} \Psi, \qquad \Pi_{a\ell}^{\mathrm{end}} = \int_{(F_\ell)_\# c_a} \Psi, \qquad \Delta_{a\ell} = \Pi_{a\ell}^{\mathrm{end}} - \Pi_a. \tag{5}$$

These operations transport and integrate the actual chain. They do not first decompose the endpoint in the original homology basis. The latter coordinates, intersections with local vanishing spheres, and monodromy matrices are not forward inputs.

Lemma 2.3. *The quantities in (5) depend only on the homology marking, the homotopy class of the loop with fixed basepoint, and the cohomology class of Ψ. They are independent of the horizontal splitting.*

Proof. The pullback family over the compact path is proper. A horizontal lift therefore exists for the whole interval and is invertible by reversed transport. The right inverses defining two splittings are joined in their affine space. Lifting this family gives an isotopy between the endpoint diffeomorphisms. The same argument over a homotopy gives homology continuation depending only on the path class. Integration of a closed form depends only on homology. □

In our family Ψ is a single-valued relative holomorphic top form, hence closed on each smooth fibre. Its value at a loop endpoint is the original Ψ, with no scalar monodromy to remove. The geometric definition above is meaningful independently of numerical access. The transport estimates below provide that access.

3. An algebraic central fibre and a transverse coefficient slice

We select algebraic coefficients within the compact eight-node construction, rather than assume that an unspecified general member has computable coefficients. We verify every central condition used by the subsequent period and analytic constructions.

3.1. **The fixed contraction.** Let $\Sigma = \mathbb{P}^1_f \times \mathbb{P}^1_g$, $L_0 = \mathcal{O}_\Sigma(1,-1)$, and use the quotient convention for

$$Z = \mathbb{P}_\Sigma(\mathcal{O} \oplus L_0) \times_\Sigma \mathbb{P}_\Sigma(\mathcal{O} \oplus L_0).$$

Write ξ_1, ξ_2 for the tautological classes and $k_i = g + \xi_i$. These classes are globally generated because their direct images are $\mathcal{O}(0,1) \oplus \mathcal{O}(1,0)$. For base coordinates s, t and local quotient coordinates (u_i, v_i), the map to A is

$$(x; y) = (v_1 s; u_1 t), \qquad (x'; y') = (v_2 s; u_2 t). \tag{6}$$

Its scheme image is the normal complete-intersection fourfold

$$V_0 = \{f_1 = x_0 x_1' - x_1 x_0' = 0, \quad f_2 = y_0 y_1' - y_1 y_0' = 0\}. \tag{7}$$

Indeed the determinant equations recover the common directions s, t whenever their corresponding vector pairs are nonzero. A direction is free exactly when both corresponding pairs vanish; both cannot be free. This proves irreducibility and the image description. The equations are a generically smooth regular sequence, so their complete intersection is reduced, with this scheme image.

The singular locus is the disjoint union

$$\ell_x = \{x = x' = 0,\ [y] = [y']\}, \qquad \ell_y = \{y = y' = 0,\ [x] = [x']\}.$$

Near either line one equation eliminates a smooth variable and the other is a 2×2 determinant in four variables. Thus its transverse singularity is an ordinary threefold node. The complete intersection is Cohen–Macaulay and regular in codimension one, which proves normality.

The map $q_Z : Z \to V_0$ contracts a ruling on each of the two sections where both quotients are onto $\mathcal{O}$ or both onto L_0. These exceptional sections have codimension two. Adjunction gives

$$-K_Z = 4g + 2\xi_1 + 2\xi_2 = 2k_1 + 2k_2 = q_Z^* \mathcal{O}_{V_0}(2,2), \qquad K_{V_0} = \mathcal{O}_{V_0}(-2,-2). \tag{8}$$

The contraction is small and crepant, and normality gives $q_{Z*}\mathcal{O}_Z = \mathcal{O}_{V_0}$. We use the exact polarization $H = \mathcal{O}_{V_0}(2,2)|_{X_0}$, also writing H for its first Chern class.

3.2. **Finite admissibility conditions.** Let $\mathcal{H} = H^0(A, \mathcal{O}(2,2)) \cong \mathbb{C}^{100}$, and set $X_h = V_0 \cap \{h = 0\}$ and $Y_h = q_Z^{-1} X_h$. The map $\mathcal{H} \to H^0(Z, -K_Z)$ is surjective: the twisted Koszul resolution

$$0 \longrightarrow \mathcal{O}_A \longrightarrow \mathcal{O}_A(1,1)^{\oplus 2} \longrightarrow I_{V_0}(2,2) \longrightarrow 0$$

has no higher cohomology in its first two terms. Normality and projection formula then give the assertion. The target has dimension $100 - 2 \cdot 16 + 1 = 69$.

The fixed toric fan has rays

$$e_1, -e_1, e_2, -e_2, f_0, e_1 + e_2 - f_0, g_0, -e_1 - e_2 - g_0. \tag{9}$$

Its sixteen maximal cones choose one ray from each consecutive pair and are unimodular. Integer linear algebra supplies the sixteen affine charts and anticanonical trivializations. The full torus support can be computed directly by substituting

$$(x_0, x_1, y_0, y_1) = (as, a, t, 1), \qquad (x_0', x_1', y_0', y_1') = (bs, b, t, 1) \tag{10}$$

in the standard $(2,2)$ monomials. It has 69 distinct monomials. This avoids a choice of sign for the polar-polytope convention.

Define $\mathcal{H}^{\mathrm{adm}}$ by the following conditions.

(1) In each toric chart, the polynomial H_α representing $q_Z^* h$ has no common zero with its four partial derivatives.
(2) On each of ℓ_x, ℓ_y, the restricted binary quartic has nonzero discriminant and nonzero values at the two fixed endpoints.
(3) All coefficients in the 69-monomial support are nonzero. For every face Θ, including the full polytope, the face polynomial H_Θ has no common zero in $(\mathbb{C}^*)^4$ with $s\partial_s H_\Theta, t\partial_t H_\Theta, a\partial_a H_\Theta, b\partial_b H_\Theta$.

These are nonempty Zariski-open conditions over $\mathbb{Q}$. The first is openness of smoothness in a projective divisor family; the second is a finite set of polynomial nonvanishing conditions. The third is the face-nondegeneracy condition for the fixed Newton polytope. The general nondegenerate anticanonical construction in [W26a, Appendix A] satisfies smoothness and the quartic conditions. Surjectivity of the section map permits the additional nonzero coefficient conditions. Each open condition is dense in the irreducible space $\mathcal{H}$, so their finite intersection is nonempty.

Proposition 3.1. *Membership in $\mathcal{H}^{\mathrm{adm}}$ is decidable for represented algebraic coefficients. Enumerating $\mathbb{Q}(\sqrt{-1})^{100}$ and testing membership therefore terminates with a coefficient vector $h_* \in \mathcal{H}^{\mathrm{adm}}$.*

Proof. For the first condition, test whether each ideal generated by H_α and its derivatives contains one. For the third, clear Laurent monomials and adjoin $z\,stab - 1$, with a new variable z; the ideal contains one exactly when the torus zero set is empty. Gröbner reduction over the represented number field and the Nullstellensatz decide these tests. Quartic discriminants, endpoint values, support coefficients and polytope faces require finite exact algebra as well. Every candidate is decided. The rational complex coefficient vectors are Zariski dense, so one lies in the open set. □

3.3. **Central invariants and attachment.** For every admissible h, the divisor Y_h is smooth. Away from the two singular lines q_Z is an isomorphism, so X_h has no singularities there. At a simple quartic zero, eliminate the coordinate along its line; the transverse determinant becomes exactly a threefold node. Thus X_h has eight nodes, in two groups of four, and $Y_h \to X_h$ is a projective small crepant resolution. Its exceptional curves have

normal bundle $\mathcal{O}(-1)\oplus\mathcal{O}(-1)$: on the all-$\mathcal{O}$ section the normal bundle in Z is $(L_0^{-1})^{\oplus 2}$, and on the all-L_0 section it is $L_0^{\oplus 2}$; the simple quartic derivative removes the trivial direction normal to the ruling. In divisor coordinates (f,g,ξ_1,ξ_2), their degree vectors are

$$\Gamma_f=(1,0,0,0),\qquad \Gamma_g=(0,1,-1,-1). \tag{11}$$

We state the fixed global toric data used here. A nondegenerate anticanonical member in this smooth toric refinement is simply connected and has $(h^{1,1},h^{2,1})=(4,60)$; its four indicated divisor classes span rational H^2. The Hodge-number calculation is [W26a, Appendix A], and the lattice-quotient verification of simple connectivity is [W26b, Section 8]. These assertions apply to every admissible member: nondegeneracy is included above, and the smooth proper family over the connected open set $\mathcal{H}^{\mathrm{adm}}$ also transports the topology from a general member. No additional Picard-rank genericity is assumed.

For clarity, the relevant intersection calculation is

$$\left(\int_{Y_h} D_iD_jH\right)_{i,j}=\begin{pmatrix}0&8&4&4\\8&0&12&12\\4&12&-8&0\\4&12&0&-8\end{pmatrix},\qquad (D_i)=(f,g,\xi_1,\xi_2). \tag{12}$$

It follows by integrating $D_iD_j(-K_Z)^2$ using $f^2=g^2=0$, $\xi_i^2=(f-g)\xi_i$, and $\int_Z fg\xi_1\xi_2=1$. Its determinant is -16384 and $H^3=320$. Thus the four classes are independent, and (11) has rank two.

Delete disjoint node balls to obtain the common exterior U. Thom and Mayer–Vietoris give, on a nearby smooth fibre,

$$H^2(Y_h;\mathbb{Z})\xrightarrow{r}\mathbb{Z}^8\xrightarrow{\partial}H^3(X;\mathbb{Z}),\qquad r(D)_i=D\cdot C_i,\quad \partial e_i=\mathrm{PD}_L(\delta_i). \tag{13}$$

Indeed the exceptional curves have real codimension four, so $H^2(Y_h)\to H^2(U)$ is an isomorphism. A resolution cap retracts to S^2 and its link restriction records the curve degree; a smoothing cap retracts to S^3 and has no degree-two cohomology. The connecting image of its link generator is the oriented Thom dual of that sphere. These local descriptions identify the maps and give exactness, with the left duality convention used below.

The restrictions of f,g are the two block indicators $v_f,v_g\in\mathbb{Z}^8$. The rational image of r is their span by (11) and the spanning divisor classes. Every integral vector in this rational span is constant with an integer value on each block; hence $\operatorname{im} r=\mathbb{Z}v_f\oplus\mathbb{Z}v_g$. It follows that the eight local classes have precisely these integral relations and rank-six span. The following integral argument also determines the saturation of their image.

Proposition 3.2. *For every admissible h, one has $H_2(Y_h;\mathbb{Z})\cong\mathbb{Z}^4$ and $H^3(Y_h;\mathbb{Z})$ is torsion-free. The vanishing lattice $L=\langle\delta_1,\dots,\delta_8\rangle\subset H_3(X;\mathbb{Z})$ is primitive, and*

$$H_3(X;\mathbb{Z})/L\cong\mathbb{Z}^{128}.$$

Proof. The very ample bundle $\mathcal{O}(2,2)$ embeds V_0 so that X_h is a hyperplane section. Its complement is affine, and

$$q_Z: Z\setminus Y_h\longrightarrow V_0\setminus X_h$$

is proper, being the base change of the projective contraction. Its fibres are points off the two singular lines and projective lines over their complements of the four quartic zeros. If $\phi(k)$ is the complex dimension of the locus of fibres of dimension k, then $\phi(0)=4$ and $\phi(1)=1$, with no other nonempty fibre loci. The homotopy-dimension theorem of Goresky–MacPherson

[GM88, Part II, §1.1*, p. 152], applied to the proper map into the affine hyperplane complement, gives a CW complex of real dimension at most $\max_k(2k+\phi(k)) = 4$ homotopy equivalent to $Z \setminus Y_h$. Integral duality on the compact real eight-manifold Z gives

$$H_k(Z, Y_h; \mathbb{Z}) \cong H^{8-k}(Z \setminus Y_h; \mathbb{Z}) = 0 \qquad (k \le 3).$$

The pair sequence therefore identifies $H_2(Y_h;\mathbb{Z})$ with $H_2(Z;\mathbb{Z}) = \mathbb{Z}^4$, the latter by the projective-bundle formula. Universal coefficients now imply that $H^3(Y_h;\mathbb{Z})$ is torsion-free.

Write $N_i \cong S^2 \times S^3$ for the node links, $B_i \simeq S^3$ for the smoothing caps, and $B_i' \simeq S^2$ for the resolution caps. On the smoothing side, Mayer–Vietoris continues (13) as

$$\mathbb{Z}^8 \xrightarrow{\partial} H^3(X;\mathbb{Z}) \longrightarrow H^3(U;\mathbb{Z}) \oplus \bigoplus_i H^3(B_i;\mathbb{Z}) \longrightarrow \bigoplus_i H^3(N_i;\mathbb{Z}).$$

Each cap-to-link restriction in degree three is an isomorphism. The kernel of the last arrow consequently projects isomorphically onto $H^3(U;\mathbb{Z})$, so $\operatorname{coker}\partial \cong H^3(U;\mathbb{Z})$. On the resolution side, the degree-two restrictions $H^2(B_i';\mathbb{Z}) \to H^2(N_i;\mathbb{Z})$ are isomorphisms. Hence the corresponding sequence gives

$$0 \longrightarrow H^3(Y_h;\mathbb{Z}) \longrightarrow H^3(U;\mathbb{Z}) \longrightarrow \bigoplus_i H^3(N_i;\mathbb{Z}) = \mathbb{Z}^8.$$

Thus $H^3(U;\mathbb{Z})$ is an extension of torsion-free groups and is torsion-free. Poincaré duality identifies it with $H_3(X;\mathbb{Z})/L$, whose rank is $134 - 6 = 128$. □

Orient the spheres accordingly and put

$$R = \left\{ p \in \mathbb{C}^8 : \sum_{i \in I_f} p_i = \sum_{i \in I_g} p_i = 0 \right\}. \tag{14}$$

The toric anticanonical sequence gives $K_{Y_h} = \mathcal{O}$ and $H^1(\mathcal{O}_{Y_h}) = H^2(\mathcal{O}_{Y_h}) = 0$. Rationality of the nodes descends these vanishings to X_h, and adjunction gives $\omega_{X_h} = \mathcal{O}$.

We use the following precise deformation consequence from [W26c, Section 2] together with the relative embedding in [W26d, Section 2]. For this fixed small transition there is a proper polarized smoothing over a full ball in R, with smooth total space and a nowhere-zero relative three-form Ω, whose oriented local periods are its coordinates. Its local equations are $\sum_{j=1}^4 w_j^2 = p_i/u_i(p)$, with holomorphic units u_i; its full discriminant is $\bigcup_i \{p_i = 0\}$. The family embeds relatively in A with the three fixed bidegrees and exact polarization $\mathcal{O}(2,2)$. These results require the Calabi–Yau vanishings, projective small resolution, exceptional rank-two map, and full-support smoothing relation just verified. They follow from unobstructed polarized deformation, the local-to-global smoothing map, the attachment sequence, and extension of the two ambient line bundles. The conclusion is an existence theorem; we do not assume that its arbitrary holomorphic embedding data are evaluated.

3.4. **Rank and algebraic first derivatives of the period map.** Fix the selected h_* and $c_* = (f_1, f_2, h_*) \in \mathcal{C}$. On a sufficiently small coefficient neighbourhood, the complete intersections are flat and Gorenstein. Fix the standard algebraic trivialization of

$$K_A \otimes \det(\mathcal{O}(1,1)^{\oplus 2} \oplus \mathcal{O}(2,2)) = \mathcal{O}_A$$

and the order of the equations. Adjunction gives a single relative residue generator Ψ, including at nodal fibres. Local Morse coordinates and the sphere integral give holomorphic

periods

$$P_i(c) = \tau_i(c)u_i(c), \qquad u_i(c_*) \neq 0. \tag{15}$$

The attachment relations hold on the common exterior and its nearby fillings. They imply $P(c) \in R$ on smooth fibres, and then everywhere by holomorphic continuation.

Apply the preceding existence theorem to this selected central member. Its relative embedding supplies a holomorphic coefficient map j from a ball in R with $j(0) = c_*$. The quotient of $j^*\Psi$ by Ω is a holomorphic unit on the total space. Properness, connected fibres and $H^0(\mathcal{O}_{X_p}) = \mathbb{C}$ imply $\pi_*\mathcal{O} = \mathcal{O}$, so this quotient is a unit $a(p)$ on the base. Consequently

$$P(j(p)) = a(p)p, \qquad DP_{c_*} \circ Dj_0 = a(0)I_R. \tag{16}$$

Thus DP_{c_*} has rank six. Only existence of j was used; no numerical inversion of that particular map is required.

The node coordinates are algebraic quartic roots and can be isolated. At each node choose a nonzero two-equation Jacobian minor and eliminate its two variables. The first two jets of the remaining four-variable equation are algebraic. Its central Hessian is nonsingular; symmetric elimination and algebraic square roots put its quadratic part into $\sum w_j^2$. The critical value derivative $\mathrm{d}\tau_i(c_*)$ is therefore algebraic. If $a_i \neq 0$ is the algebraic central residue coefficient in these coordinates, scaling the three-sphere gives

$$\mathrm{d}P_i(c_*) = \epsilon_i \pi^2 a_i \, \mathrm{d}\tau_i(c_*), \qquad \epsilon_i \in \{1, -1\}. \tag{17}$$

The unit's leading term uses only the linear Jacobian and residue value; higher Morse terms do not contribute because the period contains the factor τ_i. The full bounded version of this calculation is proved in Section 4.

Test the finitely many signs in (17) for which the two block row sums vanish, after dividing by π^2. These are algebraic equality tests. The attachment relations and rank six show that a compatible choice exists and that its two block rows span the entire left kernel. Other successful choices differ by a common sign on each block. This uses only the known source relations, not coordinates of any local sphere in the global middle lattice.

Retain the first three rows in each block to identify R with $\mathbb{C}^6$. The resulting 6×132 derivative has a nonzero six-column minor. Removing its common power of π^2 makes selection of such a minor an exact algebraic operation. The six coordinate directions define an affine six-plane through c_*, with algebraic coefficients at algebraic parameter values. This is the slice required in Theorem 4.2. That theorem supplies an evaluated inverse on a known full period ball and a smooth algebraic reference fibre. No ratio restriction on its nonzero periods has been introduced.

4. Local periods and an effective inverse neighbourhood

We construct local period coordinates with quantitative evaluation data. The central coefficients and the six transverse coefficient directions are algebraic. The inverse period section will generally have transcendental coefficients. We therefore distinguish exact algebraic operations from the evaluation of analytic functions throughout the construction.

4.1. **Evaluation data.** A name for a complex number supplies rational complex approximations with any prescribed positive absolute error. Statements about an arbitrary named parameter are relative to this access; they do not assert that every complex number is computable. For computable parameters the procedures below are ordinary algorithms.

Definition 4.1. A bounded analytic presentation of a holomorphic function consists of a polydisc with specified positive rational radii, an evaluation procedure, and a known finite upper bound on a slightly larger closed polydisc. All derivative bounds are taken on specified strictly smaller domains.

The larger complex domain is essential. Cauchy's formula both bounds derivatives and evaluates them by contour quadrature. We do not infer derivative access from an arbitrary pointwise evaluator.

Let c_* be an algebraic coefficient triple defining the eight-node central complete intersection in $\mathbb{P}^3 \times \mathbb{P}^3$. Fix six algebraic coefficient directions for which the local period derivative has rank six, and write $s \in \mathbb{C}^6$ for their coordinates. The resulting ordered triple of equations $F(s)$ has bidegrees $(1,1),(1,1),(2,2)$. Let Ψ be its ordered relative residue form. The node labels and oriented block relations are those of Section 3; retain its space R in (14). The map $\pi_R : R \longrightarrow \mathbb{C}^6$ retains the first three coordinates in each block. Its inverse fills the fourth coordinate with the negative sum of the other three.

Theorem 4.2. *From these algebraic data one can construct a positive rational radius ρ, a ball $B_{\mathrm{ev}} = \{p \in R : \|p\| < \rho\}$, and a holomorphic coefficient map σ with the following properties. The fibres*

$$X_p = Z\big(F(\sigma(\pi_R p))\big)$$

have local periods exactly p_i for the form Ψ_p. Their discriminant is exactly $\bigcup_i \{p_i = 0\}$, and every singularity of a fibre is one of the prescribed ordinary double points. The family is flat and projective, its total space is smooth, its polarization is $\mathcal{O}(2,2)$, and its relative dualizing form is Ψ. The coefficient functions, local normal forms, residue units and derivatives of every prescribed finite order have bounded analytic presentations on specified smaller domains.

One can also select an algebraic coefficient point s_* whose fibre is smooth and whose evaluated period point p_* belongs to $B^\circ_{\mathrm{ev}} = B_{\mathrm{ev}} \setminus \bigcup_i \{p_i = 0\}$, with $s_* = \sigma(\pi_R p_*)$.

No comparison between the nonzero coordinates p_i occurs in this statement. The theorem supplies a known observation domain; it does not give a complexity bound as a period approaches zero.

4.2. **An implicit function with bounds.** We first give the analytic operation used in the proof. Suppose that $G(y,z)$ has a bounded analytic presentation, $G(0,0) = 0$, and $A = D_y G(0,0)$ is invertible, with an evaluated inverse and a known upper bound for its norm. Use the maximum norm and its induced operator norm. Bounds on the second derivatives allow us to find positive rational r, η such that, for $\|y\| \le r$ and $\|z\| \le \eta$,

$$\|I - A^{-1} D_y G(y,z)\| \le q = \tfrac{1}{4}, \qquad \|A^{-1} G(0,z)\| \le r/4. \tag{18}$$

One may first require strict stronger bounds, so their verification does not require an equality test on a named complex number. The map $T_z(y) = y - A^{-1}G(y,z)$ sends the closed r-ball into the $r/2$-ball and is a q-contraction. Its unique fixed point satisfies

$$\|y(z) - T_z^n(0)\| \le \frac{q^n}{1-q}\,\frac{r}{4}. \tag{19}$$

Choose n to control this tail, then evaluate the finitely many compositions with errors whose propagated sum is smaller than the remaining tolerance. The interior margin keeps these compositions in their specified domains.

Computable local inversion with derivative access is classical; see [Zie06, Theorem 13 in the arXiv-v2 text]. Here the displayed contraction estimates provide the additional complex-domain bounds used by the subsequent period construction.

Uniform convergence of the holomorphic iterates proves holomorphy of $y(z)$ and gives the bound $\|y\| \le r/2$. Cauchy's formula on an interior product contour evaluates any prescribed derivative. A bound on a larger interior contour controls the variation of the integrand, and therefore the error of its Riemann sums. This proves both derivative evaluation and derivative bounds. Sums, products, compositions with a known image margin, and reciprocals with a known positive denominator bound preserve these properties.

4.3. **Parameter-preserving holomorphic Morse coordinates.** At an algebraic central node, choose a nonzero two-equation Jacobian minor. The preceding construction eliminates two affine variables, leaving a bounded holomorphic equation $f(z,s) = 0$ in four variables z. Its central Hessian is nonsingular. A second implicit solution, now of $D_z f = 0$, gives the critical point $c(s)$. Set $\tau(s) = -f(c(s), s)$. Taylor's formula gives

$$\begin{aligned} f(c(s)+v,s) - f(c(s),s) &= v^T H(v,s) v, \\ H(v,s) &= \int_0^1 (1-t) D_z^2 f(c(s)+tv, s)\, \mathrm{d}t. \end{aligned} \tag{20}$$

The integral is evaluated on a fixed real interval using a bound on the next derivative. Thus H is a bounded evaluated holomorphic symmetric matrix. Its central value is algebraic and nonsingular. Symmetric elimination over the algebraic numbers gives R_0 with $R_0^T H(0,0) R_0 = I$.

Writing $v = R_0 u$, put $K(u,s) = R_0^T H(R_0 u, s) R_0 - I$. On a known smaller domain $\|K\| \le 1/4$. The series

$$C(u,s) = \sum_{j \ge 0} \binom{1/2}{j} K(u,s)^j \tag{21}$$

has remainder at most $q^{N+1}/(1-q)$ after degree N, where $q = 1/4$. Each power of the symmetric matrix K is symmetric, and the convergent series identity gives $C^T C = C^2 = I + K$. Hence $u^T (I+K) u = (Cu)^T (Cu)$. The map $u \mapsto C(u,s)u$ has derivative I at the central point. Applying (18) once more supplies an inverse with the parameter s unchanged. In the resulting coordinates the equation is

$$Q(w) = w_1^2 + w_2^2 + w_3^2 + w_4^2 = \tau(s). \tag{22}$$

All maps and inverses have known domains and derivative bounds. For the convergent analytic splitting lemma without these evaluation data, see [GP26, Theorem 2.4 in the arXiv-v2 text].

The ordered complete-intersection residue becomes

$$\Psi_s = a(w,s) \operatorname{Res} \frac{\mathrm{d}w_1 \wedge \cdots \wedge \mathrm{d}w_4}{Q(w) - \tau(s)}. \tag{23}$$

The first elimination divides by the chosen equation minor; subsequent changes multiply by evaluated Jacobian determinants. The minor is bounded away from zero on a smaller domain. Thus a has a bounded analytic presentation and $a(0,0)$ is nonzero and algebraic. The fixed local orientation can be included in the sign of a.

4.4. **A period series valid at the discriminant.** Choose a known rational $r > 0$ and a parameter polydisc such that a is holomorphic near their product with $\|w\|_\infty \le r$, and

$|a| \le M$. Let α_0 be half the standard oriented volume form on the unit sphere S^3, so $\int_{S^3} \alpha_0 = \pi^2$. The function

$$A(q,s) = \int_{S^3} a(qy,s)\alpha_0(y)$$

is holomorphic and even in q: the antipodal map of S^3 has degree $+1$. Consequently

$$A(q,s) = \sum_{k\ge 0} b_{2k}(s) q^{2k}, \qquad |b_{2k}(s)| \le \pi^2 M r^{-2k}. \tag{24}$$

The bound is Cauchy's estimate on the q-circle.

These coefficients can be evaluated without integrating a shrinking sphere. Write $a(w,s) = \sum_\alpha a_\alpha(s) w^\alpha$. Odd sphere moments vanish by reflection. Separating four Gaussian integrals and then using polar coordinates gives, for $|m| = k$,

$$\int_{S^3} y^{2m}\alpha_0 = \frac{\prod_{j=1}^4 \Gamma(m_j + 1/2)}{\Gamma(k+2)} = \frac{\pi^2}{(k+1)!}\prod_{j=1}^4 \frac{(2m_j)!}{4^{m_j} m_j!}.$$

It follows that

$$b_{2k}(s) = \frac{\pi^2}{(k+1)!}\sum_{|m|=k} a_{2m}(s)\prod_{j=1}^4 \frac{(2m_j)!}{4^{m_j} m_j!}. \tag{25}$$

Each coefficient a_{2m} is a derivative divided by its factorial and is evaluated by the Cauchy procedure above.

Define $U(t,s) = \sum_{k\ge 0} b_{2k}(s) t^k$. For $|t| \le r^2/4$,

$$\left| U(t,s) - \sum_{k=0}^{N} b_{2k}(s) t^k \right| \le \pi^2 M \frac{(|t|/r^2)^{N+1}}{1 - |t|/r^2} \le \frac{4}{3}\pi^2 M\, 4^{-(N+1)}. \tag{26}$$

For $\tau \ne 0$, parametrizing the local sphere by $w = qy$, $q^2 = \tau$, multiplies the standard residue by q^2. The local period therefore has the holomorphic continuation

$$P(s) = \tau(s) U(\tau(s), s). \tag{27}$$

Either square root gives the same integral. Formula (27) evaluates the period even at $\tau = 0$; it neither tests whether a named number is zero nor divides by τ.

Since $U(0,0) = \pi^2 a(0,0) \ne 0$, evaluation of its nonzero central value and derivative bounds give a further known shrink on which

$$0 < c \le |U(\tau(s), s)| \le C. \tag{28}$$

Repeating the construction at eight nodes and taking finitely many minima supplies one common coefficient neighbourhood, with the same evaluation properties for all the periods and their derivatives.

4.5. **Excluding other singularities.** Choose disjoint coordinate boxes about the eight algebraic nodes, with algebraic centres and rational side lengths. Their closures lie inside the elimination and inverse Morse domains. Shrink the boxes so that every common zero in each closure belongs to the uniquely eliminated branch; the strict implicit-function margins preserve this property under a known coefficient change.

On the complement of these boxes in the compact ambient product use the fixed Fubini–Study metrics and connections. If $A_0 = \nabla_x F(0)$, then

$$g_0(x) = |F(0,x)|^2 + \lambda_{\min}(A_0 A_0^*)$$

is strictly positive there. A zero would be a singular common zero outside the eight boxes. A positive rational lower bound m is found by successive finite meshes and known variation bounds. The central compact domains and coefficients are algebraic, so their coverage conditions can also be checked by real algebraic quantifier elimination. Polynomial dependence on s and the eigenvalue perturbation inequality give a known coefficient radius on which $g_s \geq m/2$. Thus no new singularities occur on that complement. In each node box, (22) shows that a singularity occurs precisely when $\tau_i = 0$ and is then an ordinary double point.

At every fibre point the three equations form a regular sequence. At a smooth point this follows from their independent differentials; at a node two equations eliminate variables and the third is the nonzero quadratic hypersurface equation. The fixed-degree Koszul resolution gives a constant Hilbert polynomial and flatness over the coefficient neighbourhood. Adjunction gives the relative form Ψ. The same Koszul resolution gives $H^0(\mathcal{O}_{X_s}) = \mathbb{C}$ and $H^1(\mathcal{O}_{X_s}) = H^2(\mathcal{O}_{X_s}) = 0$. The fibres are normal, since they are local complete intersections with at most isolated nodes. Connected normal fibres are integral. The line bundle $\mathcal{O}(2,2)$ remains relatively ample.

4.6. **Period inversion and an algebraic reference fibre.** The two sphere relations imply $P(s) \in R$. Set $Q_0(s) = \pi_R P(s)$. Its central derivative is the prescribed invertible matrix, with entries in $\pi^2\overline{\mathbb{Q}}$. Apply the bounded implicit construction to $Q_0(s) - v = 0$. It gives a holomorphic $s = \sigma(v)$ on a known neighbourhood, with

$$P(\sigma(\pi_R p)) = p. \tag{29}$$

Choose a rational ball $B_{\mathrm{ev}} \subset R$ whose image under π_R lies strictly inside that neighbourhood and whose image under $\sigma\pi_R$ lies strictly inside the geometric coefficient domain. The computed image bounds give this choice. Equations (27)–(28) show that

$$\tau_i(\sigma(\pi_R p)) = p_i/U_i(p), \qquad 0 < c \leq |U_i(p)| \leq C.$$

This proves the exact discriminant assertion. At a nodal point $\mathrm{d}\tau_i = U_i^{-1}\mathrm{d}p_i$; the coordinate functional p_i is nonzero on R. The total space is therefore smooth there, and it is smooth over every regular fibre point as well.

There is also a known small coefficient ball on which $\sigma(Q_0(s)) = s$ and $P(s) \in B_{\mathrm{ev}}$: shrink until the derivative bound for Q_0 puts its image inside the inverse domain, and use uniqueness of the implicit solution. Enumerate algebraic points in this ball. The smooth locus is nonempty and open, because the period map is locally invertible and no coordinate p_i vanishes identically on R. Algebraic points are dense. At an algebraic candidate, smoothness is decided by the projective polynomial Jacobian rank tests. The search therefore finds a smooth s_*. The period $p_* = P(s_*)$ is evaluated by the series above and satisfies the assertions of Theorem 4.2.

The reference fibre has algebraic equations, although its period point need not be algebraic. This is the fibre on which we use exact semialgebraic topology. At general period points we use analytic evaluations with strict margins, without applying algebraic sign tests to transcendental coefficients.

5. An integral marking by smooth singular cycles

The reference fibre in Theorem 4.2 is a smooth complete intersection with algebraic coefficients. We now construct an integral marking on that fibre, together with smooth representatives suitable for integration. The construction uses the global fibre, but does not express any local vanishing sphere in the resulting basis.

Theorem 5.1. *Let $X \subset \mathbb{P}^3 \times \mathbb{P}^3$ be a smooth complete intersection of bidegrees $(1,1)$, $(1,1)$ and $(2,2)$ with represented algebraic complex coefficients. There is a terminating procedure producing an integral symplectic basis of $H_3(X;\mathbb{Z})$ and, for each basis element, a finite integral sum of smooth singular three-simplices in X. The face cancellations are exact. Each simplex map has a specified neighbourhood of the closed simplex on which it is smooth; its values and derivatives through order two are evaluated with known finite bounds.*

The real and imaginary parts of an algebraic coefficient are supplied with their polynomial and isolating data. Exact real algebraic decisions below apply to this representation. They are not decisions about arbitrary computable real numbers.

5.1. **A compact real algebraic presentation.** Embed each factor $\mathbb{P}^3$ in the real vector space of Hermitian matrices by its rank-one orthogonal projector. It is described by $P^* = P$, $P^2 = P$, and $\operatorname{tr} P = 1$. In the product embedding, the squared norm of a homogeneous equation on unit representatives is a polynomial in the two projector matrices: expand the squared modulus and replace $z_i\overline{z}_j$ by P_{ij} in each factor. The three squared equations therefore give a compact real algebraic presentation of X.

These squared equations need not have independent differentials. For tangent spaces we retain the original complex equations in projective affine charts. Their real derivative has rank six on X. Its kernel, carried into the projector embedding by the algebraic chart map, describes TX effectively on a finite cover.

The compatible semialgebraic triangulation theorem gives a finite simplicial complex K and a semialgebraic homeomorphism $h : |K| \longrightarrow X$, with algebraic descriptions of its graphs on simplices [BPR06, Theorem 5.43 and Remark 11.46(b)]. Here and below the locators for this reference are those of its 2016 posted revision. We use the classical construction of a map into X, rather than a simplicial replacement known only to have the same homology. No differentiability of h is assumed.

5.2. **An effective normal tube.** We give the required retraction construction for a compact smooth real algebraic submanifold $M \subset \mathbb{R}^N$ with effective local regular equations. Its tangent descriptions give a semialgebraic normal bundle

$$NM = \{(x, v) : x \in M,\ v \perp T_x M\}.$$

The map $E(x, v) = x + v$ has invertible differential along the zero section. There exists $r_0 > 0$ such that it is locally invertible and injective on $\|v\| < r_0$. For injectivity, otherwise equal-image pairs with normal lengths tending to zero would, after passing to a subsequence, have the same limiting basepoint in the compact manifold. Both pairs would then lie in one inverse neighbourhood of its zero normal vector, a contradiction. Compactness also makes local invertibility uniform near the zero section.

For a positive rational r, the assertions that E is injective on $NM_{<r}$ and that its differential is invertible there are first-order conditions over the real algebraic numbers. Injectivity uses two normal vectors and equality of their images. Differential invertibility is checked in the finite semialgebraic normal-bundle charts; their derivative graphs are obtained by implicit differentiation. Real quantifier elimination decides these conditions. Testing $r = 2^{-k}$ therefore terminates.

The image of the accepted tube is exactly $T_r = \{y : \operatorname{dist}(y, M) < r\}$. A nearest point exists by compactness, and the difference from a nearest point is normal. Injectivity of E proves that this point is unique for $y \in T_r$ and also for every image $E(x, v)$ with $\|v\| < r$.

The first component of E^{-1} is consequently a smooth nearest-point retraction

$$r_M : T_r \longrightarrow M. \tag{30}$$

Its graph is semialgebraic. This proves the particular retraction we need; general semialgebraic tubular-neighbourhood existence results alone would not supply its radius and derivative bounds.

On the compact closed tube of radius $3r/4$, every fixed derivative of r_M is bounded. Its graph is effectively semialgebraic, either by implicit differentiation in inverse charts or by the first-order definition of derivatives followed by quantifier elimination. Enumerating integer component bounds and deciding the universal inequalities produces finite bounds of any prescribed order.

At an algebraic point in T_r, the unique value of r_M, and of each derivative, is algebraic and can be isolated. At a named point y in the smaller tube of radius $r/2$, choose an algebraic q with $\|q - y\| < r/8$. The joining segment lies in the tube of radius $3r/4$. A bound for one further derivative controls the error made by replacing y by q. Refining q and the algebraic isolation therefore evaluates $r_M(y)$ and its required derivatives. No sign or equality test on y is used.

5.3. **Face-compatible smoothing.** Order the vertices of K once. The graph of h on a standard closed d-simplex has algebraic data, so values at algebraic points are algebraic. It also has an effective uniform modulus of continuity: for any rational $\epsilon > 0$, search for a rational $\delta > 0$ such that

$$\|u - v\| \le \delta \quad \Longrightarrow \quad \|h(u) - h(v)\| \le \epsilon \qquad (u, v \in \Delta^d). \tag{31}$$

This is a first-order algebraic condition on the graph. Uniform continuity proves termination. Take minima over the finite complex, and obtain a common finite bound $R_0 \ge 1$ for $\|h\|$ in the same way.

For barycentric coordinates $\lambda_0, \dots, \lambda_d$, use one common degree $n \ge 1$ and the Bernstein polynomial

$$B_n h(\lambda) = \sum_{|\alpha|=n} \frac{n!}{\alpha!} \lambda^\alpha h(\alpha/n). \tag{32}$$

Its coefficients are algebraic. On a face all terms involving an omitted vertex vanish; the remaining terms give exactly the same polynomial as the construction on that face. The restrictions of the maps thus agree as parametrized maps, not just as sets.

We record a uniform approximation bound. In the coordinates $(\lambda_1, \dots, \lambda_d)$, let A have the multinomial distribution with parameters n, λ. Then $B_n h$ is the expectation of $h(A/n)$, and

$$\mathbb{E}\,\|(A_1/n, \dots, A_d/n) - (\lambda_1, \dots, \lambda_d)\|^2 \le d/(4n).$$

Choose δ from (31) for $\epsilon/2$. Splitting at distance δ and applying Markov's inequality gives

$$\|B_n h - h\|_\infty \le \epsilon/2 + R_0 d/(2n\delta^2). \tag{33}$$

One common integer $n \ge \max_d R_0 d/(\epsilon\delta^2)$ therefore gives error at most ϵ on the entire complex.

Apply the normal-tube construction to $M = X$ and choose $\epsilon < r/8$. Define, simplex by simplex,

$$g = r_X \circ B_n h. \tag{34}$$

It has the same exact face identities. It is smooth on a known neighbourhood of each closed simplex: bound the polynomial derivatives on a larger compact box and choose a sufficiently small neighbourhood whose image stays in $T_{r/2}$. If L bounds its first derivative, one may take a radius smaller than the box margin and $3r/[8(1+L)]$. The chain rule gives the evaluated bounds

$$\begin{aligned} \|Dg\| &\le \|Dr_X\|\,\|D(B_n h)\|, \\ \|D^2 g\| &\le \|D^2 r_X\|\,\|D(B_n h)\|^2 + \|Dr_X\|\,\|D^2(B_n h)\|. \end{aligned} \tag{35}$$

The face-compatible homotopy

$$(u,t) \longmapsto r_X\big((1-t)h(u) + tB_n h(u)\big)$$

is defined because its argument stays within ϵ of $h(u)$. It identifies g_* with h_* on integral homology. We assert neither injectivity nor immersion for g. It produces smooth singular chains and does not require a C^2 triangulation theorem.

5.4. **Integral homology and orientation.** The integer boundary matrices of the ordered finite complex compute integral homology and cohomology by Smith reduction, including chain representatives. The smooth-CICY topology gives $H_1(X;\mathbb{Z}) = 0$, $H_2(X;\mathbb{Z}) = \mathbb{Z}^2$, and free $H_3(X;\mathbb{Z})$ of rank 134. Thus its middle intersection form is unimodular and alternating. In particular these computations retain integral indices; rational linear algebra would not suffice.

It remains to select the complex orientation. Choose either primitive generator z_6 of $H_6(K;\mathbb{Z})$. For the fixed product Fubini–Study form ω_K, put $\beta = (\omega_K|_X)^3/3!$. Since g is homotopic to the homeomorphism h,

$$\int_{g_\# z_6} \beta = \pm \operatorname{Vol}_{\omega_K}(X) \neq 0.$$

This sign is determined by integration with an absolute error bound. If C_1, C_2 bound the first two derivatives of a six-simplex map and W_0, W_1 bound a smooth ambient extension of β and its first derivative, its pullback integrand has Lipschitz bound

$$W_1 C_1^7 + 6 W_0 C_1^5 C_2. \tag{36}$$

Subdivision and pointwise evaluation consequently give arbitrarily small absolute enclosures after summing with the absolute chain coefficients. Since the true integral is nonzero, an enclosure eventually excludes zero. Reverse z_6 if necessary. This determines the complex orientation using only the known ambient positive form.

5.5. **Intersection form and symplectic basis.** Let $\xi_1, \dots, \xi_{134}$ be an integral cohomology basis computed on K. The simplicial cup product gives

$$Q_{ab} = \langle \xi_a \smile \xi_b, z_6 \rangle.$$

Use the left cap convention specified by

$$\langle \eta, \xi \cap z_6 \rangle = \langle \xi \smile \eta, z_6 \rangle. \tag{37}$$

Then $c_a = \xi_a \cap z_6$ are integral three-cycles whose classes are a basis by Poincaré duality. Their oriented homological intersection matrix is Q in this convention. The cup/cap identities and integral duality are those of [Hat02, Theorem 3.30 and Proposition 3.38]; the explicit order in (37) fixes the sign in degree three.

Integral symplectic reduction finds $S \in \mathrm{GL}_{134}(\mathbb{Z})$ with $S^T Q S$ equal to the chosen standard symplectic matrix. To see termination directly, a primitive vector has a primitive pairing row

by unimodularity, so Bézout's identity finds a partner pairing to one. Their hyperbolic plane splits integrally; the orthogonal projection uses integer pairing values. Compute a basis of the complement by Smith reduction and repeat.

Apply these same integer combinations to the cycles c_a, and then apply $g_\#$. Exact face identities preserve every chain boundary cancellation. Equations (30)–(35) give the asserted domains, evaluations and bounds for their smooth simplex maps. This proves Theorem 5.1.

The resulting marking is constructed from the global fibre equations. The period observations will identify the local vanishing spheres in this marking.

6. Represented transport and absolute period errors

We now construct the geometric observations used in Section 7. The domain is the known ball B_{ev} of Theorem 4.2. The reference point p_* has algebraic fibre equations, and Theorem 5.1 supplies its marking by smooth integral chains.

Theorem 6.1. *For every named $p \in B_{\mathrm{ev}}^\circ$ one can construct a geometric symplectic marking of X_p by evaluated smooth chains, and eight evaluated smooth based loops, each a positive meridian of one discriminant hyperplane and with zero winding about the others. The initial and endpoint periods of every marked chain along these loops have arbitrarily accurate absolute enclosures obtained by finite computation relative to the name of p.*

The construction uses the fibre equations, their local residue form, and the independently constructed marking. It uses no local-sphere coordinate column, intersection vector, endpoint homology decomposition, or monodromy matrix.

6.1. **Paths with known margins.** Let $x, y \in B_{\mathrm{ev}}^\circ$ be named points. Enumerate $q \in R \cap (\mathbb{Q} + \sqrt{-1}\mathbb{Q})^8 \cap B_{\mathrm{ev}}$ until the segments $[x, q]$ and $[q, y]$ avoid every coordinate hyperplane. Such q form a dense open set. For a fixed nonzero x_i, the segment from x_i to q_i contains zero precisely when q_i belongs to the closed opposite ray $\{-ax_i : a \geq 0\}$. Its inverse image under the nonzero complex-linear coordinate functional on R has empty interior. The finitely many conditions for x and y therefore exclude only a closed nowhere-dense set.

The search is effective by dovetailing candidates and refinements. At a candidate, a slope bound for each scalar segment comes from endpoint enclosures. Values on a finite time mesh, less evaluation errors and the slope bound times the cell diameter, give lower bounds for its modulus. If the segment avoids zero, sufficiently fine meshes prove a positive rational lower bound. Bad candidates need not be decided. Also find positive lower bounds for $\rho - \|x\|, \rho - \|y\|, \rho - \|q\|$. Convexity of the ball gives their minimum as an interior margin for the polygonal path.

Fix once the smooth increasing function

$$\theta(t) = \frac{\int_0^t b(u)\,\mathrm{d}u}{\int_0^1 b(u)\,\mathrm{d}u}, \qquad b(u) = \begin{cases} \exp[-1/(u(1-u))], & 0 < u < 1, \\ 0, & \text{otherwise.} \end{cases} \tag{38}$$

Extend θ by zero and one outside the interval. Every positive-order derivative vanishes at its endpoints. Values and bounds for any fixed number of derivatives are computable: inside the interval the derivatives are rational functions times the exponential; exponential decay controls their endpoint tails, and compact interior intervals have elementary bounds. Quadrature evaluates the integral, whose denominator has a positive lower bound from any fixed interior subinterval.

Parametrize each segment with θ and concatenate. This gives a smooth path with known derivative bounds and the same positive margins. In particular it constructs a path from p_* to p. Its choice need not be canonical.

6.2. **Individual meridians and flat local spheres.** For node i, choose $j \neq i$ in the same block and put $v_i = e_i - e_j \in R$. Choose a rational complex point a_i with $(a_i)_i = 0$, all other coordinates nonzero, and $\|a_i\| < \rho/4$. Such points exist because no other coordinate hyperplane contains $\{p_i = 0\} \subset R$. A sufficiently small rational $r_i > 0$ satisfies

$$\|a_i\| + r_i\|v_i\| < \rho/2, \qquad r_i|(v_i)_k| < \tfrac{1}{2}|(a_i)_k| \quad (k \neq i). \tag{39}$$

The disc $a_i + \zeta v_i$, $|\zeta| \leq r_i$, meets just the ith hyperplane, transversely at its centre. Its boundary $a_i + r_i e^{2\pi\sqrt{-1}t} v_i$ has winding one about that hyperplane and zero about the others. Join its starting point to p by the preceding path construction, follow the circle, and return along the reverse path. Flat reparametrization at the joins gives the required smooth based loop ℓ_i, with known derivative bounds, a positive lower bound for every $|p_k|$, and a known margin inside the ball.

The local Milnor homology at a threefold node is generated by the oriented real S^3 in $\sum w_j^2 = \tau_i$. One circuit in τ_i changes w to $-w$, whose action on S^3 has degree $+1$. Thus the generator defines a single-valued flat local section. Inclusion into the compact fibre commutes with transport, so the classes δ_i are globally flat on B°_{ev}. Their representatives occupy disjoint Milnor balls and have zero pairwise intersections.

On the transverse disc above, $\tau_i = \zeta / U_i(a_i + \zeta v_i)$ has a simple zero and winding $+1$. The total space over the disc is smooth and its central fibre has just one ordinary double point. This is precisely the one-critical-point hypothesis of the positive dimension-three Picard–Lefschetz formula [Del73, Exposé XIV, §3.2.3 and Theorem 3.2.11]. Changing the whisker conjugates its action by a symplectic monodromy fixing δ_i, and hence leaves that action unchanged. The individual observation uses the full six-dimensional base: varying p_i alone would violate its block relation.

6.3. **A horizontal field determined by the equations.** Compose a constructed path with $\sigma\pi_R$. Its known compact domain margin and the bounded analytic presentation of σ give evaluations and bounds for every fixed number of derivatives of the defining sections. Use the fixed Fubini–Study metrics and connections on A and on the equation bundle $\mathcal{E} = \mathcal{O}(1,1)^{\oplus 2} \oplus \mathcal{O}(2,2)$. Regard this as a real rank-six bundle for adjoints. If $F(t)$ denotes the three sections and $A_F = \nabla_x F$, then

$$g(t,x) = |F(t,x)|^2 + \lambda_{\min}(A_F A_F^*) > 0 \qquad ((t,x) \in [0,1] \times A). \tag{40}$$

A zero would be a common zero with deficient equation derivative, contrary to smoothness. Compactness makes the lower bound positive.

A rational $m > 0$ with $g \geq m$ is found without solving for the fibre points. On finitely many compact ambient coordinate cells, evaluate $F, A_F A_F^*$ and their variation bounds. The inequality $|\lambda_{\min}(B) - \lambda_{\min}(C)| \leq \|B - C\|$ controls the least eigenvalue between mesh points. Refinement of the mesh and evaluation tolerances eventually proves a positive common lower bound. This is a strict positive-margin search on the entire compact ambient space.

On $|F|^2 \leq m/2$ we have $A_F A_F^* \geq m/2$. Choose a smooth cutoff χ equal to one on $[0, m/8]$ and zero on $[m/4, \infty)$. The field

$$V = -\chi(|F|^2) A_F^* (A_F A_F^*)^{-1} \nabla_t F \tag{41}$$

on $|F|^2 < m/2$, extended by zero elsewhere, is smooth because its support has a buffer inside the invertibility region. On the zero set $\nabla_t F + A_F V = 0$. Thus (∂_t, V) is tangent to the total zero manifold, and its flow transports one actual fibre to the next. Compactness gives existence for the whole interval, and reverse transport gives the inverse.

Every derivative bound follows from the equation bounds and m. On the support the inverse has norm at most $2/m$, and

$$D(B^{-1}) = -B^{-1}(DB)B^{-1}.$$

Three derivatives of F suffice for the second spatial derivative of V. Higher jet evaluation below uses correspondingly higher derivatives, supplied by the bounded analytic presentation.

Embed A by its two rank-one Hermitian projectors in a Euclidean space. A fixed tubular retraction is obtained near each projector by taking the separated top eigenspace projector. Its uniform eigenvalue gap gives derivative bounds. Compose V with this retraction and a compact cutoff equal to one near A, to obtain a globally defined Euclidean field $U(t,x)$ with bounded derivatives. It agrees with V on A. Its solutions starting there stay there by uniqueness.

6.4. **Transported derivatives and their evaluation.** For an initial simplex map f write $x(t,s) = \Phi_t(f(s))$, $P = D_s x$, and $Q = D_s^2 x$. They satisfy

$$\dot{x} = U(t,x), \qquad \dot{P} = D_x U(t,x)P, \qquad \dot{Q} = D_x U(t,x)Q + D_x^2 U(t,x)[P,P]. \tag{42}$$

Let L bound $\|D_x U\|$, M_2 bound $\|D_x^2 U\|$, and F_1, F_2 bound the corresponding initial simplex derivatives on their given neighbourhoods. Then

$$\|P\| \le B_1 = e^L \max(1, F_1), \qquad \|Q\| \le B_2 = e^L(F_2 + M_2 B_1^2). \tag{43}$$

The first estimate is Gronwall's inequality; the second follows by variation of constants in the last equation.

These equations evaluate both derivatives, not just the map. On $\|P\| \le B_1 + 1$, $\|Q\| \le B_2 + 1$, their right side has a known Lipschitz bound, using $D_x^3 U$ in the last component. Append smooth cutoffs in P, Q which are one there and vanish outside larger bounded sets. The resulting augmented field $\mathcal{F}$ is globally bounded and Lipschitz, and the true jet solution stays where the cutoffs equal one.

For completeness, suppose $\|\mathcal{F}\| \le M$, $\|D_z \mathcal{F}\| \le L_*$ and $\|\partial_t \mathcal{F}\| \le M_t$, enlarging L_* to be at least one. Its exact solutions have $\|z''\| \le M_t + L_* M$. Euler steps of length $h = 1/N$, with evaluated field errors at most ϵ and initial error at most e_0, obey

$$\|z_N - z(1)\| \le e^{L_*} e_0 + \frac{e^{L_*} - 1}{L_*} \left(\epsilon + \tfrac{1}{2}(M_t + L_* M)h\right). \tag{44}$$

This follows by summing the one-step defect and using $(1 + hL_*)^N \le e^{L_*}$. Refining N, ϵ and e_0 therefore evaluates the transported map and its two derivatives at every named simplex point to any requested accuracy.

The actual maps remain smooth on the original simplex neighbourhoods, with bounds (43). They have exact matching face maps because the same actual flow is applied to the same initial face maps. Their numerical enclosures are not asserted to constitute a separate exactly closed chain. Applying this transport from p_* to p gives the geometric symplectic marking at the target fibre.

6.5. **A first-jet error estimate for integration.** Fix one target cycle $c = \sum_\nu m_\nu (f_\nu)_\# [\Delta^3]$ and a meridian at p. Let $\bar{\Psi}_p$ be a smooth ambient extension of the three-form on the endpoint fibre. Such an extension with evaluated bounds can be constructed from regular residue charts. Choose nonzero three-equation minors on a finite cover of the smooth fibre; the compact positivity bound (40), strict minor inequalities and refined coordinate cells supply an effective cover and denominator bounds. Local residue expressions extend near those cells. A computable smooth partition of unity and the ambient tube extend the form to a

compactly supported Euclidean form agreeing on tangent vectors to X_p. This yields known bounds $W_0 = \sup \|\bar{\Psi}_p\|$, $W_1 = \sup \|D\bar{\Psi}_p\|$.

Use common F_1, F_2 for the finitely many simplex maps, and the constants in (43). Put

$$\begin{aligned} S &= \sum_\nu |m_\nu| \operatorname{vol}(\Delta^3), \\ C_q &= W_1 B_1^4 + 3W_0 B_1^2 B_2, \\ C_j &= W_1(B_1+1)^3 + 3W_0(B_1+1)^2, \\ \Lambda_* &= 1 + 2L + M_2(B_1+1). \end{aligned} \tag{45}$$

All norms are Euclidean operator norms, with absolute values for complex coefficients. A zero chain needs no numerical integration.

Proposition 6.2. *At each quadrature point suppose a first-jet approximation $\widehat{z} = (\widehat{x}, \widehat{P})$ is absolutely continuous and piecewise C^1, has initial error at most e_0, satisfies $\|\widehat{P}\| \le B_1 + 1$, and has differential-equation residual at most r in the block maximum norm. Define*

$$E = e^{\Lambda_*} e_0 + \frac{e^{\Lambda_*} - 1}{\Lambda_*} r < 1.$$

For a simplex mesh of maximal diameter h and scalar integrand-evaluation error at most ξ, the resulting endpoint sum $\widehat{I}$ satisfies

$$\left| \widehat{I} - \int_{(\Phi_1)_\# c} \Psi_p \right| \le S(C_q h + C_j E + \xi). \tag{46}$$

Proof. For $z = (x, P)$ the right side of the first two jet equations is

$$\mathcal{F}_1(t, z) = (U(t, x), D_x U(t, x) P).$$

On the convex set $\|P\| \le B_1 + 1$ it has Lipschitz constant at most $L + M_2(B_1 + 1) \le \Lambda_*$. Subtract the exact and approximate integral equations and apply Gronwall. This gives $\sup_t \|z - \widehat{z}\|_* \le E$.

The actual endpoint integrand is $G(s) = \bar{\Psi}_{p, x(1,s)}(P_1, P_2, P_3)$. Differentiating in a unit simplex direction produces one form-coefficient term bounded by $W_1 B_1^4$ and three derivative-vector terms with total bound $3W_0 B_1^2 B_2$. Thus G has Lipschitz constant C_q, and the quadrature error on a simplex is at most its volume times $C_q h$.

At a sample point, first change the point of evaluation of the form, then change its three arguments one at a time. The first-jet bound and $\|P\|, \|\widehat{P}\| \le B_1 + 1$ give error at most $C_j E$. Scalar evaluation adds ξ. Summing with exact cell volumes and absolute chain coefficients gives (46). □

An approximate $\widehat{P}$ need not be the derivative of a global approximate chain. It approximates the actual first jet at one integration point, and the displayed estimate compares it with the actual transported chain.

6.6. **Finite refinement and the observation budget.** We justify that every positive integral tolerance is attained. First-jet solutions are C^1 on the compact time interval and have $\|P\| \le B_1$. Rational piecewise polynomial curves are dense in C^1 there. Consequently approximations exist with arbitrarily small initial errors and residuals, and with a strict margin inside $\|\widehat{P}\| < B_1 + 1$.

Enumerate such curves and finite time meshes, dovetailing the tests. Initial enclosures, polynomial derivatives and the known variation bounds for $\mathcal{F}_1$ give upper bounds for the initial error, residual and derivative-block norm on every mesh cell. Accept only after the prescribed strict inequalities are verified. A sufficiently accurate polynomial approximation has a strict margin, so finite refinement eventually verifies one. This proves termination of a residual-controlled calculation, without presuming an error certificate for an unspecified solver. There are only finitely many sample points at each simplex subdivision.

For a requested endpoint error $\eta > 0$ on a nonzero chain choose

$$h < \frac{\eta}{3S\max(1, C_q)}, \qquad \xi < \frac{\eta}{3S}, \qquad E < \min\left(\frac{1}{2}, \frac{\eta}{3S\max(1, C_j)}\right).$$

The preceding search achieves the required E by taking $e_0 < E/(2e^{\Lambda_*})$ and $r < E\Lambda_*/[2(e^{\Lambda_*} - 1)]$. All choices can use rational tolerances and upper enclosures of the constants. Identity transport gives the initial integral. Evaluate initial and endpoint periods to error $\eta/2$ each; their difference has absolute error at most η, irrespective of cancellation.

The local period is $t_i = p_i$ and is evaluated from the given coordinate name, or equivalently from the branch-free local series of Section 4. Refining the 1072 complex jumps and eight local periods supplies all observations needed by Lemma 7.1. This proves Theorem 6.1.

Every construction is on the actual compact fibre. The horizontal field uses an auxiliary ambient metric and equation Jacobians; it does not require a Ricci-flat metric, a Hermitian–Einstein metric, or an eigenform evaluator. For every fixed smooth target and chosen loop its constants are finite and its refinement terminates. No uniform running time or uniform internal precision is asserted as the loop approaches the discriminant.

7. Recovery of integral cycles from period jumps

We give the finite-accuracy argument independently of the construction of the observations. Let $\Lambda = H_3(X_p; \mathbb{Z})$ and let $\gamma = (\gamma_1, \dots, \gamma_{134})$ be an integral marking. Its intersection matrix J is alternating and unimodular. Write $\delta_i = \gamma d_i$ for the oriented local vanishing spheres and set

$$D = (d_1 \ \cdots \ d_8) : \mathbb{Z}^8 \longrightarrow \Lambda, \qquad n_i = Jd_i, \qquad t_i = \int_{\delta_i} \Psi_p = p_i. \tag{47}$$

The two block relations are known; the columns of D are the unknowns. Their image is primitive by Proposition 3.2; the recovery procedure below does not require that fact as an input.

In the positive Picard–Lefschetz convention in complex dimension three, [Del73, Exposé XIV, Theorem 3.2.11], or directly in homology [LPPV24, equation (8) in the arXiv-v2 text], transport around a positive individual node meridian gives

$$T_i\gamma_a = \gamma_a + \langle \gamma_a, \delta_i\rangle \delta_i, \qquad \Delta_{ai} = (n_i)_a t_i, \qquad T_i = I - d_i d_i^T J. \tag{48}$$

Here Δ_{ai} is the endpoint period minus the initial period of a transported geometric representative of γ_a. Its forward evaluation does not require the homology coordinates of that endpoint. The inversion of (48) is the classical rank-one Picard–Lefschetz argument; for recovery of homological monodromy from continued periods see [LPPV24, §3.5.2, equation (38), arXiv-v2 text]. The issue addressed here is its use with absolute error in both the numerator and the nonzero denominator.

7.1. **A strict stopping criterion.**

Lemma 7.1. *Suppose $\Delta = nt$ with $n \in \mathbb{Z}$ and $t \neq 0$. Let $\widehat{\Delta}, \widehat{t}$ satisfy*

$$|\widehat{\Delta} - \Delta| \leq e_\Delta, \qquad |\widehat{t} - t| \leq e.$$

If $|\widehat{t}| > e$, define

$$H_{\mathrm{obs}} = \frac{|\widehat{\Delta}| + e_\Delta}{|\widehat{t}| - e}, \qquad r_{\mathrm{obs}} = \frac{e_\Delta + H_{\mathrm{obs}} e}{|\widehat{t}|}. \tag{49}$$

If $r_{\mathrm{obs}} < 1/2$, then n is the unique nearest integer to $\mathrm{Re}(\widehat{\Delta}/\widehat{t})$. For any sequence of valid enclosures with $e, e_\Delta \to 0$, the two strict inequalities hold eventually.

Proof. The triangle inequality gives $|n| = |\Delta|/|t| \leq H_{\mathrm{obs}}$. Hence

$$\left| \frac{\widehat{\Delta}}{\widehat{t}} - n \right| = \frac{|\widehat{\Delta} - \Delta + n(t - \widehat{t})|}{|\widehat{t}|} \leq r_{\mathrm{obs}}.$$

Its real part lies strictly within $1/2$ of the integer n. As the errors tend to zero, $|\widehat{t}| \to |t| > 0$, $H_{\mathrm{obs}} \to |n|$, and $r_{\mathrm{obs}} \to 0$. □

Apply the lemma to the $134 \cdot 8 = 1072$ jumps and eight local periods. Refine all absolute enclosures and accept an entry only after the displayed strict inequalities have been verified with interval bounds. A failing strict test need not decide whether equality holds: one continues the refinement. At a fixed $p \in B^\circ_{\mathrm{ev}}$ the lemma proves finite termination for all the entries. There is no prior bound on the unknown integer heights in this procedure.

The resulting integer vectors give

$$d_i = J^{-1} n_i \in \mathbb{Z}^{134}. \tag{50}$$

Both the error in a local period and the absolute error in a small period difference are indispensable in this step. In particular, a relative error bound for each of two large endpoint integrals does not by itself control their small difference.

7.2. **Image, saturation and monodromy.** For this transition D has rank six and

$$\ker D = \mathbb{Z} v_f \oplus \mathbb{Z} v_g, \qquad v_f = \sum_{i \in I_f} e_i, \qquad v_g = \sum_{i \in I_g} e_i.$$

The identities $Dv_f = Dv_g = 0$ and $D^T J D = 0$ are useful consistency checks. They do not determine the marked inclusion D. Smith reduction of the recovered matrix gives positive integers $s_1 \mid \cdots \mid s_6$ and unimodular changes of source and target coordinates. If $L = \operatorname{im} D$ and $L^{\mathrm{sat}} = (L \otimes \mathbb{Q}) \cap \Lambda$, then

$$\Lambda / L \cong \mathbb{Z}^{128} \oplus \bigoplus_{j=1}^{6} \mathbb{Z}/s_j\mathbb{Z}, \qquad [L^{\mathrm{sat}} : L] = \prod_{j=1}^{6} s_j. \tag{51}$$

Proposition 3.2 gives $s_1 = \cdots = s_6 = 1$ and $L^{\mathrm{sat}} = L$ before any period is computed. Smith reduction independently recovers these known invariants from the observed columns. The columns also determine the position of L in the specified geometric marking.

Corollary 7.2. *Label $I_f = \{1,2,3,4\}$ and $I_g = \{5,6,7,8\}$. There is an integral symplectic basis $\alpha_1, \dots, \alpha_{67}, \beta_1, \dots, \beta_{67}$ of Λ such that*

$$\delta_i = \alpha_i \quad (1 \le i \le 3), \qquad \delta_4 = -\alpha_1 - \alpha_2 - \alpha_3,$$
$$\delta_{4+i} = \alpha_{3+i} \quad (1 \le i \le 3), \quad \delta_8 = -\alpha_4 - \alpha_5 - \alpha_6.$$

Thus the inclusion and the individual Picard–Lefschetz transformations are determined up to a simultaneous integral symplectic change of basis.

Proof. The known kernel makes $\delta_1, \delta_2, \delta_3, \delta_5, \delta_6, \delta_7$ a basis of L. Their disjoint representatives make L isotropic, and the proposition makes it primitive. Any prescribed basis $e_1, \dots, e_6$ of such a sublattice extends to a symplectic basis. Indeed primitivity allows the coordinate functional $e_j \mapsto \delta_{1j}$ to extend integrally to Λ. Unimodularity represents it by a vector f_1 with $\langle e_j, f_1 \rangle = \delta_{1j}$. The remaining e_j lie in the integral symplectic complement of the plane spanned by e_1, f_1, and span a primitive isotropic sublattice there. Induction extends the prescribed basis. The two block relations give the displayed formulas, and (48) gives the monodromy assertion. □

If D_0 denotes this normal-form matrix, then $D = S^{-1} D_0$ for some integral symplectic S. Topology determines D_0; the period observations determine D relative to the independently constructed marking, or equivalently the class of S modulo the stabilizer of D_0.

The local spheres are globally flat by Section 6, and the marking at p is transported from p_*. Their coordinate matrix is therefore constant under that transport: $D(p) = D(p_*)$. Recovery at p_* suffices to determine this matrix throughout B°_{ev}. This does not make the full middle-homology local system trivial.

Set $N_i = T_i - I$. The spheres have disjoint representatives, so $d_i^T J d_j = 0$ and therefore $N_i N_j = 0$ for all i, j. For every loop ℓ in B°_{ev},

$$T_\ell = I + \sum_{i=1}^{8} w_i(\ell) N_i, \qquad w_i(\ell) = \frac{1}{2\pi\sqrt{-1}} \int_\ell \frac{\mathrm{d}p_i}{p_i}. \tag{52}$$

Indeed fill the loop by a disc in the convex ball. Perturb its interior to meet each hyperplane transversely and to avoid pairwise intersections, which have real codimension four. Removing small discs around the finitely many inverse images expresses the boundary monodromy as a product of conjugates of individual meridians. Each local sphere is fixed by every meridian, so the conjugates have the same action on homology. The signed numbers of intersections are the windings w_i. Finally all products of two N_i vanish, which proves (52). This argument makes no assumption that the fundamental group of the hyperplane complement is abelian.

7.3. **Precision near a node.** The stopping criterion is adaptive, but a sufficient precision can also be stated in terms of the true height $H = \max_a |(n_i)_a|$. Suppose that each initial period, endpoint period, and local period is evaluated with absolute error at most $A 2^{-b}$, where $A \ge 1$. The jump error is at most $2A 2^{-b}$. The conditions

$$e_\Delta \le |t_i|/8, \qquad e \le |t_i|/[8(H+1)] \tag{53}$$

give the direct ratio error

$$\left| \frac{\widehat{\Delta}_{ai}}{\widehat{t}_i} - (n_i)_a \right| \le \frac{e_\Delta + He}{|t_i| - e} < \frac{2}{7}.$$

They also imply the strict criterion of Lemma 7.1. To see this directly, write $x = e/|t_i| \le 1/[8(H+1)]$. Then $H_{\mathrm{obs}} \le (H+1/4)/(1-2x)$ and

$$r_{\mathrm{obs}} \le \frac{1/8 + x(H+1/4)/(1-2x)}{1-x} \le \frac{2}{7} < \frac{1}{2}.$$

A sufficient common output precision is consequently

$$b \ge \left\lceil \log_2 \frac{16A(H+1)}{|t_i|} \right\rceil. \tag{54}$$

The integer height appears only in this explanatory bound, not among the inputs to the adaptive stopping rule.

For sufficiently small parameters let $\kappa_i = 2\pi/|\log|\tau_i||$. The bounded residue units give

$$|t_i| = |\tau_i U_i| \asymp \exp(-2\pi/\kappa_i).$$

Thus a sufficient number of binary output digits is

$$b \ge \frac{2\pi}{\kappa_i \log 2} + \log_2 A + \log_2(H+1) + O(1). \tag{55}$$

For direct recovery at the target fibre, this sufficient absolute tolerance is exponentially small in inverse capacity; the sufficient number of digits grows linearly in inverse capacity. Recovering the constant matrix at p_* avoids this target-dependent requirement. Bounds for transport and quadrature may deteriorate further. Equation (55) is neither a running-time bound nor a bound on the internal precision of those constructions.

7.4. **Changes of choices.** For a target basis change $\gamma' = \gamma S$ with S integral unimodular,

$$\begin{aligned} J' &= S^T J S, \quad d_i' = S^{-1} d_i, \qquad n_i' = S^T n_i, \\ \Delta_i' &= S^T \Delta_i, \quad T_i' = S^{-1} T_i S. \end{aligned} \tag{56}$$

If both bases have the same standard symplectic matrix then S is symplectic. Absolute observation errors transform with the corresponding operator norm of S^T.

Reversing the orientation of δ_i changes d_i, n_i, t_i by the same sign and leaves Δ_i, T_i unchanged. More generally, a signed permutation P of the local source basis changes D to DP and the source relations to $P^{-1}v_f, P^{-1}v_g$. In particular, the all-positive block relations must not be retained after an isolated orientation reversal.

A known meridian multiplicity $m \neq 0$ multiplies its jump by m; division by m also divides the absolute error by $|m|$. Without known winding data this primitive jump is not determined. A single-valued nowhere-zero scalar change of the relative form multiplies both the local period and the global period jump by the same scalar at the basepoint, so their ratio is unchanged. If a chosen form instead has known scalar monodromy, the endpoint must first be divided by that scalar before the initial period is subtracted. The canonical residue used here is single-valued.

7.5. **What the observations distinguish.** The marked inclusion cannot be recovered from local periods and the degree-two spectral data alone. This already follows on one fixed geometric fibre by changing its middle-dimensional marking. Choose a primitive $v \in \Lambda$ with $k = v^T J d_i \neq 0$ for one nonzero column. The integral symplectic transformations

$$S_N = I - N v v^T J, \qquad S_N^{-1} = I + N v v^T J$$

replace that column by $d_i + Nkv$. These columns are unbounded as N varies, whereas the underlying fibre, local periods, abstract intersection form and degree-two observations

remain the same. The represented global marking chains do change. Thus this argument does not produce indistinguishable inputs when the full geometric marking and all its jumps are retained.

At the scalar level adjacent numerators nt and $(n+1)t$ are separated by $|t|$; closed error discs of radius $|t|/2$ can meet. An uncertain denominator gives a further obstruction: (n,t) and $(n+1, nt/(n+1))$ have the same numerator. These examples explain the strict half-integer threshold and the role of height in denominator errors. They are scalar observation obstructions, not sharp minimax lower bounds for the full geometric inverse problem.

7.6. **The positive reconstruction theorem.**

Theorem 7.3. *Fix the algebraic central member and six coefficient directions selected in Section 3, and the resulting evaluated family over B_{ev}. For every named $p \in B_{\mathrm{ev}}^{\circ}$ there is a terminating procedure, relative to its coordinate name, which constructs a geometric symplectic marking of $H_3(X_p;\mathbb{Z})$ and recovers the marked homomorphism*

$$D : \mathbb{Z}^8 \longrightarrow H_3(X_p;\mathbb{Z}), \qquad e_i \longmapsto \delta_i.$$

It returns the primitive integral image and the individual monodromy matrices, and independently recovers the topologically determined saturation index one and quotient $\mathbb{Z}^{128}$. The monodromy matrices determine the entire homology representation by (52).

The forward observations are eight local periods and 1072 *complex jumps of marked periods around represented individual meridians. Their absolute errors are obtained from the geometric construction and refined until Lemma 7.1 applies. The two source relations and the ambient symplectic marking are known; the columns of D, their intersection vectors and the endpoint homology coordinates are not supplied to the forward calculation. For computable p this is an ordinary algorithm. Its sufficient output precision and its changes under choices are as stated above.*

Proof. Proposition 3.1 selects the central member; the rank and algebraic derivative calculation selects the slice. Theorem 4.2 supplies the known inverse period domain and an algebraic smooth reference fibre. Theorem 5.1 constructs the integral marking there without identifying any local sphere in it. Theorem 6.1 transports that marking to the target, constructs individual meridians, and supplies the absolute period enclosures by equation-based transport and integration of represented chains.

The Picard–Lefschetz identity (48) gives $\Delta_{ai} = (Jd_i)_a p_i$. Every p_i is nonzero at the target. Lemma 7.1 therefore terminates after finitely many refinements for every entry, without a height prior. Unimodularity gives the exact columns by (50); finite integer Smith operations give the remaining lattice outputs in (51). The flat-sphere argument and vanishing pairwise intersections give (52). The covariance and precision statements have been proved above. □

The theorem establishes a construction and termination result. No numerical coefficient vector, marking, period matrix or attachment matrix in the constructed marking has been calculated here. The equations determine the geometry, while the specified procedure obtains the previously unsupplied marked local inclusion from scalar observations.

8. Applicability of the analytic results

The reconstruction theorem applies on the known period ball B_{ev}. To relate its output to the analytic results of [W26c, W26d], we verify their hypotheses for the selected central member and its evaluated smoothing family. Appendix A states those results in full, including the central metrics, harmonic masses, bundle parameters and spatial norms used below.

Fix a finite graph regularity order $m \geq 4$. We shall obtain a full positive-radius ball $B_{\mathrm{an}} \subset B_{\mathrm{ev}}$, together with positive bundle-parameter and metric-basin radii. These radii may depend on the selected central coefficients and on m, and need not be computable. For $p \in B_{\mathrm{an}}^{\circ}$, write g_p for the Ricci-flat Kähler metric in $c_1(\mathcal{O}(2,2)|_{X_p})$.

Proposition 8.1. *The family of Theorem 4.2 admits B_{an} and the analytic objects specified in Appendix A, satisfying (70) and Recollections A.1–A.4. The observation procedure of Theorem 7.3 remains defined on all of B_{ev}° and in particular applies to each of these analytic target fibres.*

Proof. First, the selected central transition is linked to the original general one within the same fixed toric construction. Let $\mathcal{H}^{\mathrm{geom}}$ require only a smooth Y_h and four simple zeros on each singular line. It is a nonempty Zariski-open subset of $\mathcal{H}$, and hence path connected. It contains both the original general choice and h_*. The universal resolved divisor is smooth and proper there. Transport along a connecting path preserves the four ambient divisor classes, the exact polarization and all intersections. The four roots on each fixed line can be followed along that path, so the two exceptional types remain distinct and unchanged, up to permutations within each group. This identifies the topological inputs, not the central metrics or analytic constants.

The selected X_0 is normal Gorenstein Calabi–Yau with a projective small resolution, the required structure-sheaf vanishings, and a polarized smoothing. A line $p = tv$ with every $v_i \neq 0$ supplies the smoothing disc needed for the fixed-central conical metric theorem used in [W26c, Section 7]. Choose that metric g_0 anew. The evaluated Morse coordinates need not be its conical coordinates. The coordinate reconciliation in the same proof uses a central ambient map with $Q \circ \Phi = aQ$; replacing Φ by $a^{-1/2}\Phi$ makes it preserve Q and its nearby quadric fibres. The residual coordinate correction has positive conic order. These analytic charts coexist with the evaluated charts; the observation procedure does not evaluate the conical coordinate change.

The metric construction needs this fixed central conical metric, a smooth proper exterior, the relative polarization and form, and holomorphic quadric charts with bounded units. All are available here. Choose anew its central inverses, normalizations and finite low-growth modes. The independent-cap inverse and nonlinear correction in [W26c, Sections 7–9] then give (70) for the actual polarized Ricci-flat metrics. Their uniqueness identifies them with g_p. The Hodge trials, capacity lower bounds and mass compactness use this comparison and the same exterior restriction map. The primitive Kähler mass identity and (12), with the full surviving space removed, give exactly M_H in (58). The complete degree-one and degree-two argument in [W26c, Sections 10–12] therefore gives Recollection A.1 on these fibres. No uniform estimate over varying h follows from this reasoning.

The bundle construction requires an additional ambient fourfold condition. At c_* the first two equations form a regular sequence with normal integral zero set V_0, and h_* is a nonzerodivisor. These properties persist on an open coefficient neighbourhood: the complete-intersection family is flat, normal geometric integrality is open, and the regular-sequence conditions are open. Shrink the analytic ball so that $\sigma\pi_R$ lies in this neighbourhood. Adjunction then gives $K_{V_s} = \mathcal{O}(-2,-2)$ and makes X_s a relative anticanonical Cartier divisor. Smoothness of the triple alone was not used to assert these pair properties.

The fixed-bidegree Koszul complexes in [W26d, Section 2] now apply on this relative fourfold. Their cohomology gives the stated extension dimensions, vanishing ambient obstruction space and restriction isomorphisms. It constructs the actual relative bundle and the Chern data (74), with the two Koszul frames chosen at h_*. The required equality of the line Hilbert

polynomials follows from these same bidegrees and $H = \mathcal{O}(2,2)$, with no symmetry assumption on h_*.

The HE line background and complementary estimates in [W26d, Sections 3–5] use the actual metrics just constructed, fixed-exterior estimates, the ambient metric comparison, $H^0(L_p^{\pm 2}) = 0$, and Hartogs extension at the nodes. The nonzero exterior restriction of $\pm 2D_L$ follows from the resolution Thom sequence and $D_L^2 H = -16$; simple connectivity alone is not used to exclude off-diagonal parallel sections. The supported extension representatives come from total node trivializations and the fixed Koszul classes. Reconstruct their central elliptic solutions and positive masses $c_{\pm,0}$, and their harmonic continuations $c_\pm(p)$. These constants are not imported numerically from a different central member.

The local spin bound, critical smallness of the bounded line connection and central extension computation give the same all-degree split Dolbeault gaps. Together with the section inverse they supply the projected HE correction on a common product of base and matrix balls. The subsequent real equivariant normalization, separate orbit balancing, semistability and polystable coverage use precisely these uniform estimates and positive masses. This proves that all the hypotheses of Recollection A.2 hold here.

For flow, the required spatial heat bounds, connection expansion and contracted divergence bounds come from this same corrected family. The entry and confinement proof, invariant graph and full-data phase in [W26d, Sections 9–11] can consequently be constructed on it, with the chosen finite m. Their smallness conditions use the complementary gap, not the weakest positive finite rate. The full H^1 gauge comparison, arbitrary-unitary rigidity and actual harmonic class identification in [W26d, Sections 12–13] apply to the same connections. The finite balanced quotient and its confined paths then give Recollection A.4. Intersecting these finitely many smallness conditions leaves full positive base and bundle balls.

Finally B_{ev} retains its known radius and its observation construction. The smaller B_{an} need not contain the algebraic reference point or the chosen meridian centres. Those remain in B_{ev}°: their transports concern smooth complete intersections and the same residue form. At a target $p \in B_{\mathrm{an}}^\circ$ they produce cycles and periods on that very X_p. No PDE estimate is needed along the entire observation path, and no computation of the radius of B_{an} is required to run the procedure on B_{ev}°. □

9. Applications to Hodge spectra and HYM dynamics

We apply the reconstruction theorem to the analytic family of Proposition 8.1. The integral map recovered from periods is the vanishing-cycle map in the exact sequence underlying the Hodge comparison. We then identify the spectral, dynamical and quotient constructions of [W26d] in common coordinates: the invariant graph gives the finite Hessian, and the infinitesimal unitary action gives both the weak section energy and the vertical directions in the harmonic quotient.

Let $V = H^2(Y_{h_*}; \mathbb{R})$ and let G be its central harmonic mass. The subspace K_H of classes surviving on the smoothing has dimension two; in the fixed divisor coordinates it is spanned by $k_1 = g + \xi_1$ and $k_2 = g + \xi_2$. Represent V/K_H by $W_H = K_H^{\perp_G}$, using the G-orthogonal lifts of f, g as coordinates. Write $V_{\mathrm{lost}} = V/K_H$. The curve restriction map descends to

$$\mathsf{r} : V_{\mathrm{lost}} \longrightarrow \mathbb{R}^8, \qquad \mathsf{r}(x_f, x_g) = x_f v_f + x_g v_g. \tag{57}$$

Here v_f, v_g have the orientations fixed in (13).

Put

$$\kappa_i = \frac{2\pi}{|\log|\tau_i||}, \qquad w_\alpha = \sum_{i\in I_\alpha}\kappa_i, \qquad E_H = \begin{pmatrix} w_f & 0 \\ 0 & w_g \end{pmatrix}, \qquad M_H = \frac{4}{5}\begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}. \tag{58}$$

Here E_H comes from the capacity-weighted link restriction and M_H from harmonic mass after removing the full surviving space. Neither is defined from the small eigenvalues.

For the bundle application, set $\mathcal{W} = \mathrm{Mat}_2(\mathbb{C})^{\oplus 2}$ and $V_H = H^3/6 = 160/3$. Fix the two algebraic Koszul extension frames. Their harmonic representatives have squared masses $c_+(p), c_-(p) > 0$, uniformly bounded above and below and converging to positive central limits. These are elliptic data for g_p and its HE line backgrounds; intersection numbers do not compute them. Let $K_B = U(2)\times U(2)$ act by $(k_+,k_-)(A,B) = (k_+Ak_-^{-1}, k_-Bk_+^{-1})$. On

$$\mathfrak{h} = \{(S_+,S_-) : S_\pm = S_\pm^*,\ \operatorname{tr}S_+ + \operatorname{tr}S_- = 0\}$$

define

$$\begin{aligned} M_B(S,T) &= V_H \operatorname{tr}(ST), \\ E_B(p,z)(S) &= 2c_+\|S_+A - AS_-\|_{\mathrm{HS}}^2 + 2c_-\|S_-B - BS_+\|_{\mathrm{HS}}^2, \\ \mu_{2,p}(z) &= V_H^{-1}(c_+AA^* - c_-B^*B,\ c_-BB^* - c_+A^*A), \qquad z = (A,B). \end{aligned} \tag{59}$$

The real equivariant normalization $w = \chi_p(z)$ and section operator $\mathcal{L}_{p,w}$ are specified in Recollection A.2. The invariant graph $Z(S) = S + \psi(S)$ and split projection Π are those of Recollection A.3. On the real tangent space of $\mathcal{W}$ put

$$G_{0,p}(q,q) = 2c_+(p)\|q_A\|_{\mathrm{HS}}^2 + 2c_-(p)\|q_B\|_{\mathrm{HS}}^2. \tag{60}$$

The fundamental vector of the compact action is

$$\rho_z(\mathrm{i}S) = \mathrm{i}(S_+A - AS_-,\ S_-B - BS_+), \qquad S\in\mathfrak{h}. \tag{61}$$

The total scalar action is ineffective, so $\mathrm{i}\mathfrak{h}$ contains all the effective compact action directions.

Theorem 9.1 (Applications on the selected family)**.** *Fix the selected coefficient h_*, the evaluated family of Theorem 4.2, and a finite $m \geq 4$. On the common domain of Proposition 8.1 the following conclusions hold.*

(1) *For every $p \in B_{\mathrm{an}}^\circ$, the integral map D recovered by Theorem 7.3 is the vanishing-cycle map in the attachment sequence, and*

$$\operatorname{im}\mathsf{r} = \ker(D\otimes\mathbb{R}), \qquad \ker D = \mathbb{Z}v_f \oplus \mathbb{Z}v_g, \qquad E_H(p)(x) = \sum_i \kappa_i(p)|(\mathsf{r}x)_i|^2. \tag{62}$$

With mass M_H in (58), this energy controls the complete positive weak two-form spectrum and its observed forms as in Recollection A.1, for the actual Ricci-flat metric on this X_p.

(2) *On the same X_p, the rank-four bundles in Recollection A.2 have $c_2H = 32$ and the stated semistable and polystable types. For every sufficiently small $z \in \mu_{2,p}^{-1}(0)$ put $w = \chi_p(z)$ and fix its actual HE metric h_0. The invariant graph of Recollection A.3 defines*

$$I(S) = h_0 e^{Z(S)}, \qquad Z(S) = S + \psi(S).$$

On its actual HYM region, its finite field is the negative gradient of twice the Donaldson functional restricted to I, for the pullback of the invariant L^2 metric. At the

HE origin, in logarithmic metric coordinates,

$$\begin{aligned} J_0 &= I_{\mathfrak{h}} + D\psi(0) = (\Pi|_{S_{B,p,w}})^{-1}, \\ J_0 Df(0) &= -\mathcal{L}_{p,w} J_0, \\ g_{\mathrm{fin},0}(S,T) &= M_B(S,T) + \langle D\psi(0)S, D\psi(0)T\rangle_2. \end{aligned} \tag{63}$$

Here $S_{B,p,w}$ is the entire seven-dimensional low section space, including its kernel. If H_{fin} is this finite Hessian, then

$$|g_{\mathrm{fin},0} - M_B| \le C|z|^4 M_B, \qquad (1 - C|z|)E_B \le H_{\mathrm{fin}} \le (1 + C|z|)E_B. \tag{64}$$

In particular its kernel is exactly $\ker E_B$*, and*

$$|(H_{\mathrm{fin}} - E_B)(S,T)| \le C|z|\sqrt{E_B(S)E_B(T)}.$$

The complete low spectrum in Recollection A.2 and the all-time actual-flow comparison in Recollection A.3 consequently use these same finite metric coordinates.

(3) *The weak bundle energy is the norm of the action map:*

$$E_B(p,z)(S) = G_{0,p}(\rho_z(\mathrm{i}S), \rho_z(\mathrm{i}S)). \tag{65}$$

For genuine tangents q to a zero-level orbit-type stratum, the actual harmonic connection metric is relatively $1 + O(|z|)$ equivalent to

$$\inf_{\xi \in \mathrm{i}\mathfrak{h}} G_{0,p}(q - \rho_z\xi, q - \rho_z\xi). \tag{66}$$

Thus $\ker \rho_z$ *describes the constant infinitesimal stabilizer, whereas* $\operatorname{im} \rho_z$ *describes the vertical deformation tangents. The genuine class map, confined distances, completion and tangent cone are those of Recollection A.4.*

The comparisons are uniform over arbitrary node-capacity ratios, arbitrary positive bundle-rate ratios and independent base and bundle limits in the stated domain. Observation paths may lie in the larger B°_{ev}. The analytic radius need not be computable, and no uniformity over varying central coefficients is asserted.

The Hodge, bundle, flow and harmonic metric estimates are imported in their full stated forms from [W26c, W26d], with their hypotheses checked in Proposition 8.1. The proof below establishes their compatibility with the integral construction. It also records the spectral–flow and action-map identifications explicitly; the corresponding analytic arguments are in [W26d, Sections 10–12].

9.1. **The actual attachment sequence.** The integral Thom sequence (13) identifies the restriction of a divisor class to the ith link with its degree on the ith exceptional curve. The next map sends that link generator to the Thom class Poincaré dual to the oriented vanishing sphere. After the duality and excision identifications, it is exactly $e_i \mapsto \delta_i$. Its matrix in the geometric marking constructed here is D, because Theorem 7.3 recovers the coordinates of these spheres from their actual period jumps.

The image of the integral restriction map is $\mathbb{Z}v_f \oplus \mathbb{Z}v_g$, as computed in Section 3. Exactness gives the kernel assertion in (62). Over $\mathbb{R}$, the kernel of restriction is the full surviving two-dimensional space K_H. Its quotient therefore gives (57) and $\operatorname{im} \mathsf{r} = \ker(D \otimes \mathbb{R})$. The normalized link generator has capacity κ_i. The sum of its squared restrictions weighted by these capacities is precisely E_H. Minimizing the central mass over all representatives modulo K_H gives M_H. The actual-metric comparison then follows from Recollection A.1, including its complete kernel and complementary gap.

In particular, the two lost degree-two directions and the six-dimensional vanishing-cycle span are distinct terms of this exact sequence. The recovered placement of that span in $\mathbb{Z}^{134}$ and its individual columns add marked information beyond the known kernel of D. Its primitivity and unmarked normal form are already determined by topology. No small-spectrum measurement has been used to perform this integer recovery.

9.2. **The graph tangent and the finite Hessian.** For the space of determinant-normalized Hermitian metrics on the fixed holomorphic bundle, let

$$\mathcal{G}_h(a,b) = \int_{X_p} \operatorname{tr}(h^{-1}a\,h^{-1}b)\,\mathrm{d}V_{g_p}.$$

The HYM field $h^{-1}\dot{h} = -2\mathrm{i}\Lambda F_h$ is the negative gradient of $\mathcal{F} = 2\mathcal{D}$, where $\mathcal{D}$ is Donaldson's functional with first variation $\mathrm{d}\mathcal{D}_h(a) = \int \operatorname{tr}(\mathrm{i}\Lambda F_h\, h^{-1}a)\mathrm{d}V$. On the inner region of the invariant graph, $dI\,f = -\operatorname{grad}_{\mathcal{G}}\mathcal{F}\circ I$. For every finite tangent a this implies

$$(I^*\mathcal{G})(f,a) = \mathcal{G}(dI\,f, dI\,a) = -\mathrm{d}\mathcal{F}(dI\,a) = -\mathrm{d}(\mathcal{F}\circ I)(a).$$

This proves the finite gradient assertion on its actual domain; no such assertion is needed for the auxiliary cutoff field outside that domain.

The graph construction gives $\Pi D\psi = 0$ and $D\psi = O(|w|^2)$ in X. Its linearized invariance equation at the critical origin gives $J_0 Df(0) = -\mathcal{L}_{p,w}J_0$. The image of J_0 is in the elliptic operator domain and is invariant under the self-adjoint $\mathcal{L}_{p,w}$. The bound $Df(0) = O(|w|^2)$ puts its seven eigenvalues below the fixed complementary gap. It is therefore the whole low space. Since $\Pi J_0 = I_{\mathfrak{h}}$, J_0 is exactly the inverse restricted observer in (63).

At a critical point the second derivative of I is paired with $\mathrm{d}\mathcal{F} = 0$ in the Hessian chain rule. Consequently the exact observed forms are

$$N(S,T) = \langle J_0 S, J_0 T\rangle_2, \qquad H_{\mathrm{fin}}(S,T) = \langle \mathcal{L}_{p,w}J_0 S, J_0 T\rangle_2. \tag{67}$$

Orthogonality of $\mathfrak{h}$ and QX proves the mass identity and its relative $O(|w|^4)$ bound.

For completeness, the passage from ordered eigenvalues to the entire Hessian retains all weak directions. The primitive section estimate of [W26d, Section 6] is, with $r = |w|$,

$$(1-Cr)\{E_B(p,w)(S) + q_{p,0}(v)\} \le q_{p,w}(S+v) \le (1+Cr)\{E_B(p,w)(S) + q_{p,0}(v)\}, \tag{68}$$

for every $v \perp \mathfrak{h}$ in H^1, where $q_{p,w}(u) = \|D_{p,w}u\|_2^2$ and $q_{p,0}(v) \ge c\|v\|_2^2$. The energy notation $E_B(p,w)$ means the formula in (59) with arrows w_A, w_B. Taking $S + v = J_0 S$ gives $H_{\mathrm{fin}} \ge (1-Cr)E_B(p,w)$. The kernels agree: both correspond to the constant Hermitian stabilizer, and J_0 fixes those constant kernel directions.

Pass to the quotient by this kernel and denote the induced energies by $\bar{H}, \bar{E}$ and the quotient masses by $\bar{N}, \bar{M}$. A quotient mass is the infimum over representatives. It preserves the comparison $N = (1 + O(r^4))M_B$. The positive generalized eigenvalues of (H_{fin}, N) are the positive low eigenvalues of $\mathcal{L}_{p,w}$. Their relative comparison in the w coordinates, proved from (68), gives

$$\frac{\det \bar{H}}{\det \bar{E}} = \frac{\prod_{\lambda_j>0}\lambda_j}{\prod_{\nu_j>0}\nu_j}\frac{\det \bar{N}}{\det \bar{M}} = 1 + O(r).$$

Every eigenvalue of $\bar{E}^{-1/2}\bar{H}\bar{E}^{-1/2}$ is at least $1 - Cr$, their product is $1 + O(r)$, and their number is at most seven. Each is therefore at most $1 + Cr$. For a zero-dimensional quotient the claim is immediate. This gives the whole relative Hessian bound without a lower bound on a positive weak rate.

Finally real K_B-equivariance implies

$$\rho_w(\mathrm{i}S) = d\chi_p(z)\rho_z(\mathrm{i}S).$$

The derivative is $I + O(|z|)$ in the uniformly positive norm $G_{0,p}$. Taking its squared norm proves $E_B(p,w) = (1+O(|z|))E_B(p,z)$ as quadratic forms, with the same kernel. This proves (64). On the quotient by that kernel the normalized symmetric error has operator norm at most $C|z|$; Cauchy–Schwarz proves the stated bilinear estimate.

The phase relation is consistent with this identification. After the basin shrink in Recollection A.3, both the solution shifted by t and its tracking graph trajectory remain in the phase domain. The exponential tracking phase is unique: two finite trajectories tracking at rate c have initial separation bounded by their separation at time T times $e^{L_f T}$, which tends to zero since $L_f < c$. Thus $\Gamma(u(t)) = \phi_t\Gamma(u(0))$. On graph data $\Gamma(Z(S)) = S$. This identifies the actual evolution and its full-data phase; it does not replace that phase by the initial projection Πu_0.

9.3. **The shared action and the harmonic metric.** Equation (65) follows directly from (59) and (61). Its metric meaning requires minimizing over all spatial gauge directions. For the actual connection map $\mathcal{A}_p(w)$, the estimate proved in [W26d, Section 12] is

$$\begin{aligned}(1 - C|w|)\{G_{0,p}(q,q) + \|D_0 v\|_2^2\} &\le \|d\mathcal{A}_p(w)q - D_{\mathcal{A}_p(w)}v\|_2^2 \\ &\le (1 + C|w|)\{G_{0,p}(q,q) + \|D_0 v\|_2^2\},\end{aligned} \tag{69}$$

for every $q \in \mathcal{W}$ and complementary trace-free skew-Hermitian $v \in QH^1$. Its proof uses harmonic orthogonality of the linear connection increment to $D_0 v$, the connection remainder of order $O(|w|)$ relative to these two terms, and the uniform complementary section gap.

Infimizing first in v preserves the lower bound by $G_{0,p}$; taking $v = 0$ proves the upper bound. Every remaining gauge direction is a constant $\xi \in \mathrm{i}\mathfrak{h}$. Exact equivariance gives $d\mathcal{A}_p(w)(\rho_w\xi) = -D_{\mathcal{A}_p(w)}\xi$. Hence, after reversing the sign of ξ, apply the same inequality to every residual $q - \rho_w\xi$ and then take its infimum. The finite image of ρ_w is closed at each fixed parameter. No bound for the norm of a minimizing ξ, and no inverse of a small positive orbit singular value, is required.

The identity $d\chi_p(z)\rho_z\xi = \rho_w\xi$ transfers this comparison to z inside each residual, before minimization. For a tangent to an actual HE orbit-type stratum, the full minimum is the Coulomb representative. The linearized integrability and mean-curvature equations, together with the Kähler identities, make its $(0,1)$ part Dolbeault harmonic. Its real connection norm is twice the Dolbeault harmonic norm. This is exactly the convention used in (60) and Recollection A.4. The actual class and stabilizer assertions in that theorem then give the genuine quotient interpretation. This proves (66) through every stabilizer type.

In particular the flow graph metric acts on metrics of one fixed holomorphic bundle, while the harmonic metric acts on deformation tangents modulo $\operatorname{im}\rho_z$. Their relationship is the common action map and its stabilizer, rather than equality of the two metrics.

9.4. **Independence of the integral observations from the bundle metric.** For fixed p, w, unitary bundle gauge transformations and HYM metric evolution leave the equations of X_p, the residue form, the marked cycles and their transports unchanged. The observations $\Delta_{ai} = p_i(Jd_i)_a$ and the recovered map D are therefore independent of those bundle operations. They use the represented cycles and period enclosures of Theorem 6.1. Changes of homology marking and signed node labels instead obey the covariance and error transformations of Section 7.

The same node labels enter r, the periods p_i, and the smoothing functions $\tau_i = p_i/U_i$. Thus (55) relates the required absolute period precision to the actual capacity weights. It does not identify period error with the relative spectral error $e(p)$. Proposition 8.1 ensures that both constructions concern these same target fibres even when observation paths leave B_{an}.

9.5. **Independent base and bundle scales.**

Corollary 9.2. *Let $F(h), G(h) \to +\infty$ and set*

$$p(h) = (a, a, a, -3a;\ b, b, b, -3b), \qquad a = e^{-F(h)}, \quad b = e^{-G(h)}.$$

For all sufficiently small h these are actual smooth fibres of the selected family, with $w_f \sim 8\pi/F$ and $w_g \sim 8\pi/G$. If $F/G \to 0$, their larger and smaller positive weak Hodge rates satisfy respectively

$$\lambda_+ \sim \frac{9\pi}{8F}, \qquad \lambda_- \sim \frac{10\pi}{9G}.$$

Independently, for any $q > 1$ and $s \downarrow 0$, take

$$\sqrt{c_+(p)}\, A = \sqrt{c_-(p)}\, B = \operatorname{diag}(s, s + s^q).$$

Their actual HE images have one trace-free zero mode, two positive rates $(2/V_H)s^{2q}(1+O(s))$, and four rates asymptotic to $(8/V_H)s^2$. The actual HYM comparison holds through times of order s^{-2q} and all later times. The harmonic quotient and integral reconstruction conclusions apply on these same fibres.

Proof. The two sum-zero relations hold identically and no coordinate vanishes. The evaluated full period ball contains all sufficiently small points of this path. The residue units are bounded above and below, so the logarithms in each group differ from F or G by bounded quantities. This proves the capacity asymptotics. Formula (73), or its trace and determinant, gives the two displayed Hodge rates when $F/G \to 0$; relative actual-metric comparison preserves them.

The weighted arrows are equal diagonal matrices, so their quadratic moment map vanishes exactly. For positive diagonal entries u_1, u_2, the seven finite section rates are

$$0, \quad \frac{8u_1^2}{V_H}, \quad \frac{8u_2^2}{V_H}, \quad \frac{2(u_1 - u_2)^2}{V_H}\ \text{(twice)}, \quad \frac{2(u_1 + u_2)^2}{V_H}\ \text{(twice)}.$$

This follows by separating diagonal Hermitian entries and the real and imaginary parts of each off-diagonal entry in (59); the weighted arrow masses cancel. Substitution gives the asserted hierarchy, and Recollection A.2 gives the relative $O(s)$ actual comparison. The domains of Recollection A.3 and Theorem 9.1 impose no relation between h and s, proving the remaining assertions. □

Finite-order holomorphic arcs have a single logarithmic order, with coefficients depending on their vanishing orders. The paths in the corollary may have genuinely separated logarithmic scales and need not be holomorphic at the origin. When the ratio of the group scales has no limit, the pointwise comparisons remain valid; limiting hierarchies require subsequences or parameter sectors. The restriction and action maps above do not by themselves construct a filtered elliptic complex.

The integrated theorem is on smooth base fibres, polystable bundle parameters and the stated spatial metric basin. Singular graded concentration requires additional local data and is a separate problem. The theorem supplies neither singular-fibre HYM evolution nor a

completion of a larger moduli space. Its finite evolution is the exact corrected field, and its all-time error is absolute; a relative nonlinear decay expansion requires further estimates.

Appendix A. Analytic results used in the applications

We record the results of [W26c, W26d] with the hypotheses and normalizations used in Sections 8 and 9. The statements below are recollections of those results. Proposition 8.1 verifies their hypotheses for the algebraically selected family; Theorem 9.1 gives the resulting applications of the integral reconstruction.

A.1. **Domains and the Ricci-flat input.** Fix any finite graph regularity order $m \geq 4$. There is a full positive-radius ball $B_{\mathrm{an}} \subset B_{\mathrm{ev}}$ on which the analytic conclusions stated below hold, together with positive bundle-parameter and metric-basin radii. These radii may depend on the fixed central member and on m. No uniformity as that central member varies is asserted. Statements for an arbitrary fixed spatial derivative order retain their own order-dependent constants and neighbourhoods.

For $p \in B_{\mathrm{an}}^{\circ}$ let g_p be the Ricci-flat Kähler metric in $c_1(H)$, where $H = \mathcal{O}(2,2)|_{X_p}$. Its volume is $V_H = H^3/6 = 160/3$. The metric input in [W26c, Sections 7–10] consists of a fixed central conically singular metric g_0, conic radius functions ϱ_p, and initial Kähler metrics $\widetilde{g}_p^R$ which equal normalized Stenzel metrics on inner node regions. For every fixed spatial derivative order k, positive $\mu > \eta$ and constants C_k give

$$\limsup_{\substack{p\to 0\\ p\in B_{\mathrm{an}}^{\circ}}} \max_{j\leq k} \sup_{X_p} \varrho_p^j |\nabla^j (g_p - \widetilde{g}_p^R)|_{\widetilde{g}_p^R} \leq C_k R^{\mu-\eta}. \tag{70}$$

The order is to fix small R, let $p \to 0$, and then decrease R. On every fixed compact exterior $g_p \to g_0$ in all fixed spatial derivatives. The radius on a node chart is comparable to $(|z|^2 + |\tau_i|)^{1/3}$ and is comparable to one on the exterior. The estimates impose no ratio condition on the nonzero node parameters. They concern spatial derivatives, not differentiability with respect to p.

A.2. **The complete Hodge comparison.** Use the central harmonic mass G, surviving subspace K_H, and orthogonal representatives W_H of Section 9. On a fixed compact exterior C_* let Π_* be the central L^2 projection onto distributionally closed two-forms, with unrestricted tangential trace. The bounded exterior observer is

$$\mathcal{A}_H u = \mathrm{pr}_{W_H}^{G}[\Pi_*(u|_{C_*})]. \tag{71}$$

Exterior identifications are fixed before the degeneration. This definition applies to form-domain inputs and does not integrate an arbitrary unclosed L^2 form over cycles.

The finite energy and mass are E_H, M_H in (58).

Recollection A.1 (Ricci-flat Hodge comparison). On a sufficiently small B_{an} there are $\gamma > 0$ and $e(p) \to 0$ such that the harmonic two-form space has dimension two, with mass converging to $G|_{K_H}$, and precisely two positive two-form eigenvalues lie below γ. Their values are relatively $1 \pm e(p)$ equivalent to the generalized eigenvalues of (E_H, M_H). The first positive one-form eigenvalue and third positive two-form eigenvalue are at least γ.

The positive weak two-form space $S_{H,p}$ consists of coexact primitive real $(1,1)$ forms, and $\mathcal{A}_H : S_{H,p} \to W_H$ is an isomorphism. With $J_H = (\mathcal{A}_H|_{S_{H,p}})^{-1}$, its observed forms satisfy

$$\begin{aligned} (1-e)M_H &\leq \langle J_H \cdot, J_H \cdot \rangle_{g_p} \leq (1+e)M_H, \\ (1-e)E_H &\leq \langle \mathrm{d}J_H \cdot, \mathrm{d}J_H \cdot \rangle_{g_p} \leq (1+e)E_H. \end{aligned} \tag{72}$$

These conclusions are uniform over arbitrary node-scale ratios. The finite positive eigenvalues

$$\mu_\pm = \frac{9(w_f + w_g) \pm \sqrt{81(w_f - w_g)^2 + 4w_f w_g}}{128} \tag{73}$$

satisfy $\mu_-/\mu_+ \le 4/5$. Each actual weak eigenline is therefore identified through $\mathcal{A}_H$ with its finite generalized eigenline with error tending to zero uniformly in w_f/w_g.

This is the statement of [W26c, Theorem 1.1 and Section 12] in the present notation. The operator is the real Hodge Laplacian on its closed-manifold elliptic domain. On the coexact two-form sector its quadratic form is $\|\mathrm{d}u\|_2^2$ with H^1 form domain; the harmonic and exact sectors are accounted for separately.

A.3. **Nonflat bundles and their weak section spectrum.** On the same relative family put $L_p = \mathcal{O}(1,-1)|_{X_p}$. There is a locally free holomorphic rank-four family $\mathcal{E}_p(w)$, $w = (w_A, w_B) \in \mathcal{W} = \mathrm{Mat}_2(\mathbb{C})^{\oplus 2}$ near zero, with split member

$$\mathcal{E}_p(0) = L_p \otimes \mathbb{C}^2 \oplus L_p^{-1} \otimes \mathbb{C}^2.$$

It extends over the discriminant as an algebraic bundle construction. The metric assertions here are on smooth fibres. Writing $D_L = c_1(L_p)$, its characteristic data are

$$D_L H^2 = 0, \qquad D_L^2 H = -16, \qquad \det \mathcal{E} = \mathcal{O}, \quad c_2(\mathcal{E}) = -2D_L^2, \quad c_2(\mathcal{E})H = 32, \quad c_3(\mathcal{E}) = 0. \tag{74}$$

In particular this is a nonflat bundle family.

The local deformation germ is the full two-arrow space: $\mathrm{Ext}^0 = \mathfrak{gl}_2 \oplus \mathfrak{gl}_2$, $\mathrm{Ext}^1 = \mathcal{W}$, $\mathrm{Ext}^2 = \mathcal{W}^*$ and $\mathrm{Ext}^3 = (\mathfrak{gl}_2 \oplus \mathfrak{gl}_2)^*$. Its obstruction equations vanish by the ambient construction, despite the nonzero Ext^2 on the threefold. All small members are Gieseker semistable. Slope polystability is equivalent to a closed affine orbit of $\mathrm{GL}_2(\mathbb{C}) \times \mathrm{GL}_2(\mathbb{C})$. The small germ contains no slope-stable rank-four objects.

Use the harmonic extension masses $c_\pm$, compact group K_B, Hermitian space $\mathfrak{h}$, and finite forms and moment map in (59).

Recollection A.2 (Nonflat bundles and their section spectrum). There is a real K_B-equivariant normalization $\chi_p(z) = z + O(|z|^2)$, $d\chi_p = I + O(|z|)$, for which the transported corrected Chern connection at $w = \chi_p(z)$ has mean curvature exactly $\mu_{2,p}(z)$. Thus $\mu_{2,p}(z) = 0$ gives an HE representative. These representatives cover the sufficiently small polystable objects; this coverage uses a separate complex-orbit correction. The map χ_p is not asserted holomorphic or complex-orbit preserving.

At an HE member the trace-free Hermitian section operator $\mathcal{L}_{p,w} = D_{p,w}^* D_{p,w} = 2\bar\partial_{p,w}^* \bar\partial_{p,w}$ has exactly seven low modes, zeros included. If ν_j are the generalized eigenvalues of (E_B, M_B), in increasing order with zeros, then

$$(1 - C|z|)\nu_j \le \lambda_j(\mathcal{L}_{p,w}) \le (1 + C|z|)\nu_j \quad (1 \le j \le 7), \qquad \lambda_8(\mathcal{L}_{p,w}) \ge c > 0.$$

The kernel is the constant Hermitian stabilizer in the specified unitary transport. The full Hermitian endomorphism statement adds the identity, with a uniformly gapped nonconstant scalar sector. The operator domain is H^2 and its form domain is H^1 on each fixed smooth compact fibre.

These assertions, including the deformation and stability statements preceding this recollection, are proved in [W26d, Sections 2–7]. The four factor types, in order, are two distinct nonzero cycles, one repeated nonzero cycle, one nonzero cycle and the two zero vertex factors, and the split object. Their trace-free stabilizer dimensions are $1, 3, 2, 7$, and their positive

weak counts are $6, 4, 5, 0$. Nonzero simple cycles give stable rank-two factors with $c_2 H = 16$; zero factors are L_p and L_p^{-1}. All these types are included, with arbitrary ratios of positive weak rates and independent base and bundle limits.

A.4. **Actual flow and harmonic moduli inputs.** Fix an HE member with $w = \chi_p(z)$, and let h_0 be its HE metric. Use the specified unitary transport, the split connection D_0, and split L^2 projection Π onto $\mathfrak{h}$; put $Q = I - \Pi$. For a trace-free Hermitian logarithm define

$$\|u\|_X = \|u\|_\infty + \|D_0 u\|_\infty.$$

This is a spatial norm on metrics of one fixed holomorphic bundle.

Recollection A.3 (Hermitian–Yang–Mills metric flow). For the fixed finite $m \geq 4$, a common nonempty ball $\|u_0\|_X < \delta$ of smooth determinant-normalized metrics $h(0) = h_0 e^{u_0}$ has the following properties. There are a chosen C^m graph $Z(S) = S + \psi(S)$, a complete finite field f, and a C^1 phase Γ depending on the full spatial datum. Writing $r = |w|$, one has $\psi(0) = 0$, $\Pi\psi = 0$, $\|D^j \psi\|_X \leq C_m r^2$ for $j \leq m$ and $\|f\|_{C^m} \leq C_m r^2$. The graph is invariant for its cutoff extension and agrees with the actual HYM equation on an inner region.

For every datum in the stated ball, $S_{\mathrm{eff}}(t) = \phi_t(\Gamma(u_0))$ remains in that actual region for all $t \geq 0$, and $h_{\mathrm{eff}}(t) = h_0 e^{Z(S_{\mathrm{eff}}(t))}$ is itself an actual solution of $h^{-1}\dot{h} = -2\mathrm{i}\Lambda F_h$. With $D_* = \|Qu_0 - \psi(\Pi u_0)\|_X$,

$$\begin{aligned} \|u(t) - Z(S_{\mathrm{eff}}(t))\|_X &\leq C D_* e^{-ct}, \\ \Big\| \log\Big(e^{-Z(S_{\mathrm{eff}}(t))/2} e^{u(t)} e^{-Z(S_{\mathrm{eff}}(t))/2} \Big) \Big\|_X &\leq C D_* e^{-ct}. \end{aligned} \tag{75}$$

The corresponding metric-order bounds are $e^{-CD_* e^{-ct}} h_{\mathrm{eff}} \leq h \leq e^{CD_* e^{-ct}} h_{\mathrm{eff}}$. The phase satisfies $\Gamma(Z(S)) = S$ and, on a smaller forward-confined basin, $\Gamma(u(t)) = \phi_t \Gamma(u_0)$.

For finite preparation set $\Gamma_T = \phi_{-T}(\Pi u(T))$. For a fixed small coupling bound ϵ and $L_f = \|Df\|_\infty = O(r^2) < c$,

$$|\Gamma_T - \Gamma| + \sup_{t \geq 0} \|Z(\phi_t \Gamma_T) - Z(\phi_t \Gamma)\|_X \leq C\epsilon D_* e^{-(c - L_f)T}. \tag{76}$$

One can take $c = \sigma_0/2$, with a uniform complementary semigroup exponent $\sigma_0 > 0$. For $r > 0$, preparation time $T_r = 4\sigma_0^{-1} \log(1/r)$ gives error $O(r^2 D_*)$ and $T_r \lambda_j \to 0$ for every positive weak rate. At $r = 0$ the finite trajectory is constant.

This is [W26d, Sections 9–11]. The parameters p, w are fixed during every flow. The result is absolute tracking to the exact finite field, including all later times; it is not a relative nonlinear decay expansion.

For varying holomorphic bundle parameters, put

$$U = \mathrm{diag}(u_1, u_2), \qquad z_p(u) = (c_+^{-1/2} U, c_-^{-1/2} U), \qquad \mathsf{W} = (\mathbb{Z}/2)^2 \rtimes S_2.$$

The group acts by signs and permutation. If $\mathcal{A}_p(w)$ denotes the corrected unitary connection, define the actual class map

$$\mathfrak{F}_p([u]) = [\mathcal{A}_p(\chi_p(z_p(u)))].$$

Recollection A.4 (Harmonic moduli metric). For a common sufficiently small $R_0 > 0$, $\mathfrak{F}_p : \overline{B}_R/\mathsf{W} \to \mathcal{M}_p(R)$ is a homeomorphism onto its actual class image for $R \leq R_0$, with smooth parametrizations of its genuine orbit-type strata. On each stratum,

$$(1 - C|u|) 4|\mathrm{d}u|^2 \leq \mathfrak{F}_p^* g_{\mathrm{harm}} \leq (1 + C|u|) 4|\mathrm{d}u|^2.$$

Let $d_{p,R}$ use continuous paths confined to $\mathcal{M}_p(R)$, piecewise smooth in its finitely many strata, allowing improper endpoint lengths. Then

(77)
$$\sqrt{1-CR}\, d_{\mathrm{flat}}([u],[v]) \le d_{p,R}(\mathfrak{F}_p[u],\mathfrak{F}_p[v]) \le \sqrt{1+CR}\, d_{\mathrm{flat}}([u],[v]), \quad d_{\mathrm{flat}} = 2\min_{\gamma\in\mathsf{W}} |u-\gamma v|.$$

The open-radius locus completes to the closed-radius locus. Its fixed local intrinsic metric has unique tangent cone $(\mathbb{C}^2, 4|\mathrm{d}u|^2)/\mathsf{W}$ at the split object. The rescaled distance error is $O(\rho)$ on bounded balls, uniformly in p, also for the fixed-locus distance d_{p,R_0}.

The actual-class and full-stabilizer assertions are essential inputs here, not consequences of a formal quotient metric. They and the stated distance theorem are proved in [W26d, Sections 12–13]. Distances in a larger ambient moduli space are outside this conclusion.

References

[BPR06] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in real algebraic geometry*, second ed., Algorithms and Computation in Mathematics, vol. 10, Springer, 2006. doi:10.1007/3-540-33099-2. Theorem and page locators refer to the author-posted revision of June 26, 2016.

[Del73] P. Deligne, *Comparaison avec la théorie transcendante*, Groupes de monodromie en géométrie algébrique, II (SGA 7 II), Exposé XIV, Lecture Notes in Mathematics, vol. 340, Springer, 1973, pp. 116–164. doi:10.1007/BFb0060510.

[Ehr52] C. Ehresmann, *Les connexions infinitésimales dans un espace fibré différentiable*, Séminaire Bourbaki, vol. 1, exposé 24, 1952, pp. 153–168. NUMDAM.

[GM88] M. Goresky and R. MacPherson, *Stratified Morse theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 14, Springer, Berlin, 1988. doi:10.1007/978-3-642-71714-7.

[GP26] G.-M. Greuel and G. Pfister, *The splitting lemma in any characteristic*, J. Algebra **689** (2026), 610–628. doi:10.1016/j.jalgebra.2025.10.023. arXiv:2507.17078v2.

[Hat02] A. Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002. Author's text.

[LPPV24] P. Lairez, E. Pichon-Pharabod, and P. Vanhove, *Effective homology and periods of complex projective hypersurfaces*, Math. Comp. **93** (2024), 2985–3025. doi:10.1090/mcom/3947. arXiv:2306.05263v2.

[LS23] P. Lairez and E. C. Sertöz, *Separation of periods of quartic surfaces*, Algebra Number Theory **17** (2023), no. 10, 1753–1778. doi:10.2140/ant.2023.17.1753.

[LLW18] Y.-P. Lee, H.-W. Lin, and C.-L. Wang, *Towards A+B theory in conifold transitions for Calabi–Yau threefolds*, J. Differential Geom. **110** (2018), no. 3, 495–541. doi:10.4310/jdg/1542423628.

[Mil63] J. Milnor, *Morse theory*, Annals of Mathematics Studies, vol. 51, Princeton University Press, Princeton, 1963.

[Ser19] E. C. Sertöz, *Computing periods of hypersurfaces*, Math. Comp. **88** (2019), no. 320, 2987–3022. doi:10.1090/mcom/3430. arXiv:1803.08068v2. Section locators refer to this preprint.

[W26a] B. J. Wuebben, *Nonsimplicial local stability chambers of toric tangent bundles*, preprint, October 29, 2026. doi:10.5281/zenodo.23774770.

[W26b] B. J. Wuebben, *Integral topology and small eigenvalues in Calabi–Yau conifold transitions*, preprint, November 9, 2026. doi:10.5281/zenodo.24096887.

[W26c] B. J. Wuebben, *Relative Hodge spectra of multiscale conifold smoothings*, manuscript, November 12, 2026. doi:10.5281/zenodo.24175848.

[W26d] B. J. Wuebben, *Hermitian–Yang–Mills dynamics and harmonic moduli near a conifold*, manuscript, November 12, 2026. doi:10.5281/zenodo.24175924.

[Zie06] M. Ziegler, *Effectively open real functions*, J. Complexity **22** (2006), no. 6, 827–849. doi:10.1016/j.jco.2006.05.002. arXiv:cs/0501024v2.